

Vectors

Unit 2-2 Lecture Notes

MSDS 520: Math Fundamentals for Data Science
Meharry Medical College

These notes are the written companion to the Unit 2-2 slides and follow them section by section. They go further than the slides: every idea comes with extra **worked examples** (marked “Worked example” in bold), boxes titled **Watch out** flag the mistakes students most often make, and most sections end with extra practice. Every exercise is followed by a worked solution; try each one before reading on.

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1 Why Vectors?

1.1 Where This Shows Up in Data Science

Almost every dataset you will meet is a table: one **row** per record (a patient, a house, an image, a document) and one **column** per measurement. Here is a tiny piece of a clinic’s table, with three patients and three measurements each:

Patient	Age (years)	Systolic BP (mmHg)	A1c (%)
A	54	130	6.1
B	58	138	6.4
C	71	152	7.2

Pick a single row and read its numbers off in order, and you have a list of numbers: patient A is the list 54, 130, 6.1. That list is a **vector**, and a great deal of what data science does with a record is really something done with vectors:

- **Every record is a point.** One patient, one house, one image, or one document becomes a list of numbers, and the whole dataset becomes a cloud of points. A table of 10,000 patients with three measurements each is a cloud of 10,000 points in three-dimensional space.
- **Similarity is geometry.** Recommenders, search engines, and duplicate detection all rank things by **cosine similarity**, which is an angle. A hospital that wants to find past patients “like” a new admission can rank them this way.
- **Distance is the basis of clustering.** *k*-means, nearest neighbors, and anomaly detection all measure how far apart two vectors are. To predict how a new patient will fare, a nearest-neighbor method finds the patients whose vectors lie closest and looks at what happened to them.
- **Redundant features are dependent vectors.** When two columns carry the same information, their vectors lie on one line. That is **collinearity**, and it wrecks a regression. A table that records weight in both pounds and kilograms is the simplest example.
- **Compression is choosing a better basis.** PCA (principal component analysis), at the end of this unit, is exactly that.

Several of these words (angle, dependent, basis) will not mean much yet; each one gets its own section below. Unit 2-1 wrote systems of equations as grids of numbers. Here we ask what a single **column** of numbers means on its own.

1.2 What We Will Build

The plan, in the order we will take it:

1. A **vector**, drawn as an arrow and written as a column (or row) of numbers.
2. A vector’s **magnitude**; **adding**, **scaling**, and **subtracting** vectors; the **distance** between points; and **unit vectors**.
3. The **dot product**: one number that measures how much two vectors agree. From it come the **angle** between two vectors, **cosine similarity** (how a spam filter tells which emails are alike), **orthogonality**, and **projection**.
4. **Linear combinations**, **span**, and **independence**: which points you can reach, and whether anything is redundant.
5. **Basis**, **dimension**, **change of basis**, and a look ahead to **matrices**.

Five notebooks go with these notes:

- U2-2_Vectors-1_ArrowsAndOps.ipynb
- U2-2_Vectors-2_DotProductAndAngle.ipynb
- U2-2_Vectors-3_ProjectionAndSimilarity.ipynb
- U2-2_Vectors-4_LinearCombinationsAndSpan.ipynb
- U2-2_Vectors-5_IndependenceAndBasis.ipynb

Throughout the notes, a small line at the right margin names the notebook and section where a topic can be seen running in code.

2 Vectors as Arrows, Rows, and Columns

2.1 Two Pictures of the Same Thing

Some quantities are completely described by one number: how far, how fast, how heavy. Others also need a direction. “I walked 5 km” tells you how far I went but not where I ended up; for that you need “5 km north.”

Definition: Vector

A **vector** is a quantity with both a **magnitude** (how big) and a **direction** (which way).

Size only	Size and direction: a vector
distance: 5 km	displacement: 5 km north
speed: 250 m/s	velocity: 150 m/s east + 200 m/s north

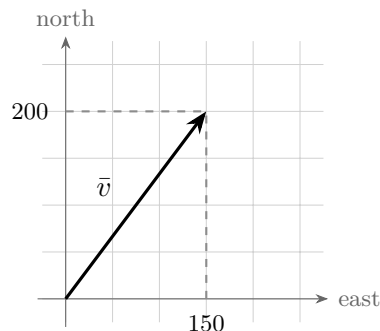
A quantity with size only is called a **scalar**; it is a single ordinary number, possibly with a unit attached. Here are more of each, so that the boundary is clear:

Scalars (one number)	Vectors (size and direction)
a patient’s temperature, 37.2°C	a wind blowing toward the east at 20 m/s
a dose of 500 mg	a push of 10 newtons to the left
a body mass of 70 kg	a walk of 3 m east and 4 m north down a ward
a plane’s speed, 250 m/s	a plane’s velocity, 150 m/s east and 200 m/s north

A temperature has no direction at all: it makes no sense to ask which way 37.2°C points. A speed does come from something moving in a direction, but the speed itself throws the direction away. That is exactly the difference between the two entries in the last row.

Take that last entry: a plane flying 150 m/s east and 200 m/s north at the same time, so that it travels diagonally, toward the northeast (a little more north than east). There are two ways to show its velocity, and they carry exactly the same information. We can **draw** it as an arrow, and we can **write** it as a list of **components**, one number per axis:

$$\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix} \begin{array}{l} \leftarrow \text{east} \\ \leftarrow \text{north} \end{array}$$

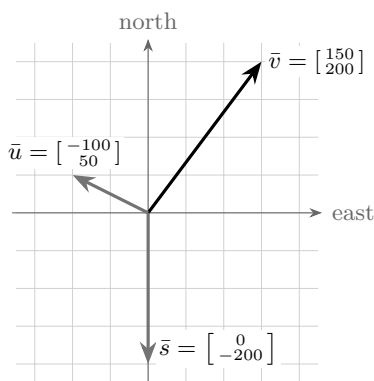


The plane's velocity in m/s;
one square is 50 m/s.

The arrow's length shows how fast the plane moves and its tilt shows which way; the two numbers say the same thing one axis at a time. Notice the scale: one square of the grid stands for 50 m/s, so the arrow runs 3 squares east and 4 squares north. (Section 3.1 shows that this plane's speed is exactly 250 m/s, which is why the table pairs that velocity with a speed of 250 m/s.)

Worked example: reading components, signs included. A component can be negative, and the sign carries direction. A second plane flying 100 m/s **west** and 50 m/s north has a negative east component, because west is “negative east.” A third, flying due south at 200 m/s, has no east component at all and a negative north component:

$$\bar{u} = \begin{bmatrix} -100 \\ 50 \end{bmatrix}, \quad \bar{s} = \begin{bmatrix} 0 \\ -200 \end{bmatrix}.$$



Three planes; one square is 50 m/s. A negative component points the opposite way along its axis.

Reading the other way, from arrow to numbers: count how far the head is from the tail along each axis, with a minus sign for west or south. The arrow $\bar{u}$ ends 2 squares west and 1 square north of where it starts, and at 50 m/s a square that is -100 and 50 .

Notation: marking a vector

- $\mathbf{v}$, $\bar{v}$, and $\vec{v}$ all mean “vector”: **bold** when typing, a **bar** or an **arrow** by hand. We use the **bar**.
- $\hat{v}$ (a **hat**) means a vector of length exactly 1, a **unit vector**.

The marker changes with the medium, never the meaning. Bold is easy to type but almost impossible to write by hand, while bars, arrows, and hats are easy to write but awkward to type in an ordinary word processor. Textbooks and software documentation mostly use bold, so expect to see all of these. These notes use the bar, to match the board, and give a unit vector a hat *instead* of a bar, never both; unit vectors arrive in Section 3.6.

Watch out: mark every vector

On paper, v and $\bar{v}$ are different objects: v looks like a single number, while $\bar{v}$ is a whole list of numbers. Writing “ $v = 250$ ” when you mean the velocity $\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$ loses the direction and makes later steps impossible to check. Put the bar on every vector, every time, and put a hat (and no bar) only on a vector you have checked has length 1.

Notebook U2-2_Vectors-1_ArrowsAndOps, §1, the arrow and the column side by side

Exercise 2.1

Scalar or vector? For each quantity, say which, and why.

1. A patient’s resting heart rate, 72 beats per minute.
2. A nurse walks 40 m down a corridor heading west.
3. An IV infusion running at 125 mL per hour.
4. Today’s wind: 15 km/h, blowing toward the south.
5. The room temperature, 21°C.

Solution to Exercise 2.1

1. **Scalar.** A rate is one number; it has no direction.
2. **Vector.** It has a size (40 m) and a direction (west). As a column with components (east, north) it is $\begin{bmatrix} -40 \\ 0 \end{bmatrix}$: west is negative east.
3. **Scalar.** The fluid goes into the patient, but “125 mL per hour” is a single amount per unit time; there is no direction in space to record.
4. **Vector.** Size 15 km/h, direction south: $\begin{bmatrix} 0 \\ -15 \end{bmatrix}$ in (east, north) components. Leave off the direction and you are left with the wind *speed*, a scalar.
5. **Scalar.** Temperature has no direction.

The test in every case is the same: does it make sense to ask “which way?” If yes, it is a vector.

2.2 Three Running Examples

Three vectors will follow us through the whole lecture. They come from very different places, and that is the point: the same mathematics handles all three.

Velocity	Patient	Email
A plane flying 150 m/s east and 200 m/s north:	Age, systolic blood pressure, A1c:	Counts of <i>free</i> , <i>win</i> , <i>cash</i> , <i>meeting</i> :
$\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$	$\bar{p} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix}$	$\bar{d}_1 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}$
One number per direction.	One number per measurement.	One number per word.

The patient’s three numbers are an age in years, a systolic blood pressure in mmHg (the upper number of a reading such as 130/85), and an A1c, a blood test of average blood sugar over the past few months reported as a percentage. This is patient A from the table in Section 1. The email vector counts how many times each of four words appears; the next subsection shows exactly how it is built.

Keep an eye on the **units**, because they decide which operations make sense later:

- **Velocity:** both components are in m/s. The two numbers are the same kind of thing, so the geometry (lengths, angles) has a real physical meaning.
- **Patient:** years, mmHg, and percent. Three different kinds of thing stacked in one list. Adding a year to a mmHg makes no sense, and Section 3.1 shows what goes wrong.
- **Email:** counts, with no units at all.

You may wonder what “direction” can mean for a patient or an email, which do not point north or east. For data, the direction of a vector is the **mix** of its components: the proportions between them. An email with twice as many of every word (counts 4, 2, 4, 0 instead of 2, 1, 2, 0) has the same mix as $\bar{d}_1$, so it points the same way, only farther. That idea, direction as proportion, is what cosine similarity will measure in Section 4.5.

The question all three build toward is one of **similarity**. Given two patients, or two emails, **how similar are they?** Is a new email **spam**? By the end of these notes, “similar” will be a number we can **compute**: either how **close** two vectors are, or how nearly they **point the same way**.

2.3 Turning an Email into a Vector

A **spam filter** decides whether an incoming email is junk, and it does so by comparing emails with one another. To compare them, it first has to turn each one into numbers.

Recipe: a count vector

1. Fix a **vocabulary**: the words to track.
2. In each email, **count** each word.
3. The list of counts **is** the vector.

Two details matter. Word order and capital letters are thrown away, so *Free* counts as *free*, and “cash free” gives the same counts as “free cash.” And real filters track thousands of words, so a real email vector has 5000 or more entries (it lives in $\mathbb{R}^{5000}$, notation we define in Section 2.6). Our

example keeps a vocabulary of just four words: *free*, *win*, *cash*, and *meeting*. Here are three emails, with the verdict a person would give each one:

$\bar{d}_1$ “Win cash now! Free cash, free entry.” **spam**
 $\bar{d}_2$ “Win cash now! Free cash, free entry. Win free cash, free prizes, cash today.” **spam**
 $\bar{d}_3$ “Free lunch at the meeting. After the meeting, free parking; bring cash.” **not spam**

Count the first one by hand. In “Win cash now! Free cash, free entry.” the word *free* appears twice, *win* once, *cash* twice, and *meeting* not at all, so its vector is 2, 1, 2, 0 read down a column. Doing the same for all three emails gives this table, in which each **column** is one email’s vector:

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
<i>free</i>	2	4	2
<i>win</i>	1	2	0
<i>cash</i>	2	4	1
<i>meeting</i>	0	0	2

Notice that every count in $\bar{d}_2$ is exactly double the count in $\bar{d}_1$. Keep that in mind; it matters a great deal in Section 4.6.

Worked example: a fourth email. Take “Win! Win! Win! Cash prize inside.” Go through the vocabulary one word at a time: *free* does not appear (0), *win* appears three times (3), *cash* once (1), and *meeting* not at all (0). So

$$\bar{d}_4 = \begin{bmatrix} 0 \\ 3 \\ 1 \\ 0 \end{bmatrix}.$$

The words *prize* and *inside* are not in the vocabulary, so they are simply ignored: the vector records only the words we chose to track.

Worked example: a clinical note. The same recipe turns any text into numbers, including the free-text notes in an electronic health record (EHR). Take the vocabulary *pain*, *chest*, *fever*, *cough* and the note “Patient reports chest pain. Pain worse on exertion; denies fever.” Counting: *pain* twice (“pain” and “Pain”), *chest* once, *fever* once, *cough* not at all:

$$\bar{n} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix} \begin{array}{l} \leftarrow \textit{pain} \\ \leftarrow \textit{chest} \\ \leftarrow \textit{fever} \\ \leftarrow \textit{cough} \end{array}$$

Watch out: counts do not see meaning

The note says the patient *denies* fever, yet *fever* gets a count of 1, exactly as if the patient had one. Counting throws away word order, and with it negation (“no,” “denies”), who did what to whom, and every other part of meaning that lives in how words are arranged. Count vectors are simple and surprisingly useful, but this blind spot is real; Section 4.7 meets a kind of vector

built to capture meaning instead.

Exercise 2.2

Using the vocabulary *free*, *win*, *cash*, *meeting*, write the count vector $\bar{e}$ of the email “Meeting at noon: free lunch, and bring cash for the free coffee.” By eye, which of $\bar{d}_1$, $\bar{d}_2$, $\bar{d}_3$ does it most resemble?

Solution to Exercise 2.2

Go word by word: *free* appears twice (“free lunch,” “free coffee”), *win* never, *cash* once, and *meeting* once (“Meeting,” with the capital thrown away). So

$$\bar{e} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

By eye it looks most like $\bar{d}_3$, the lunch-meeting email: both have no *win* and some *meeting*. Exercise 4.2 in Section 4.6 puts a number on that impression.

2.4 Columns and Rows: Two Ways to Write a Vector

The same list of numbers can be written standing up or lying down:

$$\bar{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \text{ (column, } 2 \times 1) \qquad \bar{v} = \begin{bmatrix} 3 & 4 \end{bmatrix} \text{ (row, } 1 \times 2)$$

A shape is always written as (rows) $\times$ (columns). A column of two numbers has 2 rows and 1 column, so it is 2×1 ; the same numbers in a row are 1×2 .

- The two carry the same information in **different shapes**: $n \times 1$ or $1 \times n$. For now the difference is only layout, but from Unit 2-3 on, the shape decides what can be multiplied by what.
- We write vectors as **columns by default**. A patient stored in a data table, though, is a **row**, because the table gives each patient one row.
- A **point** is neither: it stays a tuple, $(3, 4)$. The next subsection explains why a point and a vector are different ideas.

Worked example: one patient, two shapes. In the table of Section 1, patient B occupies a row, so the table stores B as a 1×3 row. When we do mathematics with B we stand it up as a 3×1 column:

$$\underbrace{\begin{bmatrix} 58 & 138 & 6.4 \end{bmatrix}}_{\text{as stored: } 1 \times 3} \qquad \bar{q} = \underbrace{\begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix}}_{\text{as we write it: } 3 \times 1}$$

Same patient, same three numbers, same order. (We call this patient $\bar{q}$ from now on; $\bar{q}$ returns in Section 3.5.) Likewise each email is a 4×1 column, and an email over a 5000-word vocabulary is a 5000×1 column.

In code, numpy (the Python library for arrays of numbers) has **three** kinds of array that can hold a vector:

Shape	Built with	What it is
$(n,)$	<code>np.array([3, 4])</code>	a flat list, no orientation
$(n,1)$	<code>np.array([[3], [4]])</code>	a column : n rows, 1 entry each
$(1,n)$	<code>np.array([[3, 4]])</code>	a row : 1 row, n entries

You can always check which one you have with `v.shape`. Most code uses the flat $(n,)$ kind, which is simply a list with no orientation at all; whether we think of it as a column or a row is up to us.

Watch out: a flat array only looks like a row

`print(np.array([3, 4]))` displays `[3 4]`, spread across the line, and it is tempting to read that as a row vector. It is not: its shape is $(2,)$, with no rows or columns at all. numpy prints a flat array across the line simply because that is how text fits on a screen. When orientation matters, check `.shape`, not the printout.

Exercise 2.3

Give the shape of each.

1. The row $\begin{bmatrix} 150 & 200 \end{bmatrix}$ and the email column $\bar{d}_1$.
2. `np.array([54, 130, 6.1])`
3. `np.array([[54], [130], [6.1]])`
4. `np.array([[54, 130, 6.1]])`

Solution to Exercise 2.3

1. The row has 1 row and 2 columns: 1×2 . The email $\bar{d}_1$ has 4 rows and 1 column: 4×1 .
2. $(3,)$: one pair of brackets, so a flat list of three numbers with no orientation.
3. $(3, 1)$: the outer brackets hold three inner lists of one number each, so 3 rows of 1 entry, a column.
4. $(1, 3)$: the outer brackets hold a single inner list of three numbers, so 1 row of 3 entries.

Counting brackets is the quick way to read a numpy shape: the outer list gives the number of rows, and each inner list gives the entries in a row.

2.5 Point Versus Vector

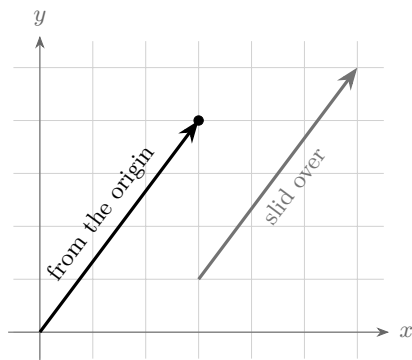
The same two numbers can describe two different ideas.

- A **point** $(3, 4)$ is a **location**. It sits where it sits.

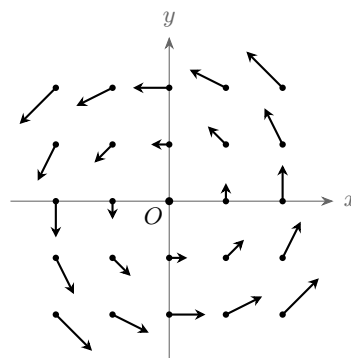
- A **vector** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is a **displacement**: 3 across, 4 up. It has **no home**.

A displacement is an instruction for how to move, “go 3 to the right and 4 up,” and that instruction makes sense no matter where you start. That has three consequences:

- We draw vectors from the origin **by default**, so the arrow ends exactly at the point with the same numbers, and arrow and point look alike.
- Sliding an arrow somewhere else, keeping the same length and direction, **changes nothing**: it is still the same vector.
- Addition, in the next section, relies on that freedom.



Two drawings of the **same** vector.
Only the dot is a point.



A **velocity field**:
fluid swirling around the origin O .

The velocity field on the right shows that freedom being used. Each arrow sits **where the fluid is** and describes how the fluid at that spot is moving, and the arrows grow longer farther out because the fluid moves faster there. Not one of them starts at the origin, and each is still a perfectly good vector.

Worked example: walking instructions on a ward. Lay a grid over a hospital floor plan, in meters. The nurses’ station is the point $(2, 1)$, room 12 is the point $(5, 5)$, and the pharmacy is the point $(-1, 3)$. These are **points**: places on the map. The walk from the station to room 12 goes $5 - 2 = 3$ m east and $5 - 1 = 4$ m north, so it is the **vector** $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$. The walk from the station to the pharmacy goes $-1 - 2 = -3$ m east (that is, 3 m west) and $3 - 1 = 2$ m north: the vector $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$. In general the displacement from the point (a_1, a_2) to the point (b_1, b_2) is

$$\begin{bmatrix} b_1 - a_1 \\ b_2 - a_2 \end{bmatrix} : \quad \text{end minus start, one coordinate at a time.}$$

Now start somewhere else. From the point $(0, 2)$ to the point $(3, 6)$ is again 3 east and 4 north, the very same vector $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, even though neither end is the station or room 12. That is what “a vector has no home” means: the instruction “3 east, 4 north” does not care where you are standing when you follow it.

Exercise 2.4

On the same floor plan, write the vector for each walk.

1. From room 12, $(5, 5)$, back to the nurses' station, $(2, 1)$.
2. From the pharmacy, $(-1, 3)$, to room 12, $(5, 5)$.
3. Is the walk from $(10, 0)$ to $(13, 4)$ the same vector as the walk from the station to room 12?

Solution to Exercise 2.4

Use end minus start in each coordinate.

1. $\begin{bmatrix} 2-5 \\ 1-5 \end{bmatrix} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$: 3 m west and 4 m south. It is the station-to-room walk reversed, so every component flips sign.
2. $\begin{bmatrix} 5-(-1) \\ 5-3 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$: 6 m east and 2 m north. Note $5 - (-1) = 6$; subtracting a negative adds.
3. $\begin{bmatrix} 13-10 \\ 4-0 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. **Yes**, the same vector: the same instruction, followed from a different starting point.

2.6 Beyond the Plane

Nothing about the definition needs exactly two components. A vector with n components lives in $\mathbb{R}^n$, the set of all lists of n real numbers:

$$\bar{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$$

Here v_1 is the first component, v_2 the second, and so on down to v_n . The symbol $\in$ means “is an element of,” $\mathbb{R}$ stands for the real numbers, and the exponent counts the components. So $\mathbb{R}^2$ is the plane and $\mathbb{R}^3$ is ordinary three-dimensional space.

The running examples live in whatever dimension their data needs:

- **Velocity:** add “up” for a plane that is climbing, and it lives in $\mathbb{R}^3$. A plane flying 150 m/s east and 200 m/s north while climbing at 10 m/s has velocity

$$\begin{bmatrix} 150 \\ 200 \\ 10 \end{bmatrix} \begin{array}{l} \leftarrow \text{east} \\ \leftarrow \text{north} \\ \leftarrow \text{up} \end{array}$$

- **Patient:** 3 measurements is $\mathbb{R}^3$; a 40-value lab panel is $\mathbb{R}^{40}$.
- **Email:** a 5000-word vocabulary puts every email in $\mathbb{R}^{5000}$.

The same counting works for any kind of data. A small grayscale image 28 pixels wide and 28 pixels tall has $28 \times 28 = 784$ pixels, each with one brightness value; list those values one after another and the image is a single vector in $\mathbb{R}^{784}$. Images of handwritten digits, a classic first dataset in machine learning, are stored exactly this way.

This raises a fair worry: you cannot **picture** $\mathbb{R}^{40}$. You do not need to. Every formula in these notes is a sum over components, so it works for **any** n exactly as it works for $n = 2$. The habit to build is this: **reason with the picture in $\mathbb{R}^2$; trust the algebra in $\mathbb{R}^n$.**

Exercise 2.5

Each item below is recorded as one vector. In which $\mathbb{R}^n$ does it live?

1. A patient's answers to a 10-question survey, each scored 1 to 5.
2. A patient's heart rate, recorded once a minute for an hour.
3. A 64×64 grayscale image.
4. The plane's position: kilometers east of the airport, kilometers north, and altitude.

Solution to Exercise 2.5

Count the numbers recorded; that count is n .

1. One number per question: $\mathbb{R}^{10}$. (The scores being whole numbers from 1 to 5 does not matter; the vector still lives in $\mathbb{R}^{10}$, it just uses only some of its points.)
2. One number per minute for 60 minutes: $\mathbb{R}^{60}$.
3. One brightness per pixel, $64 \times 64 = 4096$ of them: $\mathbb{R}^{4096}$.
4. Three numbers: $\mathbb{R}^3$. Strictly a position is a **point**, (x, y, z) , not a displacement, but its coordinates still form a list of three real numbers.

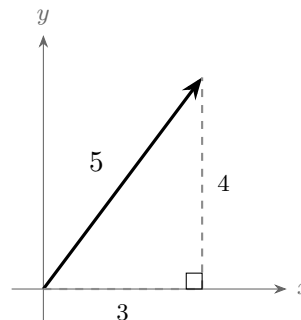
3 Vector Operations

3.1 Magnitude: How Long Is a Vector?

The first thing to ask about an arrow is how long it is. Start with the arrow $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$: it runs 3 across and 4 up, so it is the long side (the **hypotenuse**) of a right triangle whose legs are 3 and 4. The Pythagorean theorem gives its length:

$$\sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

The plane's velocity is the same triangle with every side 50 times bigger: $150 = 50 \times 3$ and $200 = 50 \times 4$, so its length is $50 \times 5 = 250$.



Legs 3 and 4, hypotenuse 5.

The same recipe (square every component, add, take the square root) works in any number of dimensions, and it is the definition of length:

Definition: Magnitude (norm)

The **magnitude**, or **norm**, of $\vec{v} \in \mathbb{R}^n$ is

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

The double bars are read “the norm of $\vec{v}$ ” or simply “the length of $\vec{v}$.” Because squares are never negative, a norm is **never negative**, and it is **zero only for $\vec{0}$** , the **zero vector** whose components are all 0: the arrow that goes nowhere. (If even one component is not zero, its square is positive, and the sum under the root is positive too.)

Applied to the running examples:

- **Velocity:** $\|\vec{v}\| = \sqrt{150^2 + 200^2} = \sqrt{22500 + 40000} = \sqrt{62500} = 250$ m/s, the plane’s **speed**. The magnitude of a velocity is a speed, which completes the table in Section 2.1.
- **Email:** $\|\vec{d}_1\| = \sqrt{4 + 1 + 4 + 0} = 3$ and $\|\vec{d}_2\| = \sqrt{16 + 4 + 16 + 0} = 6$: longer email, longer vector.
- **Patient:** $\|\vec{p}\|$ mixes years, mmHg, and percent, so it is **meaningless**. Squaring an age and a blood pressure and adding them adds quantities measured in different units, and the result measures nothing.

Procedure: computing a magnitude

1. Square every component (a negative component becomes positive).
2. Add the squares.
3. Take the square root of the total.

Worked example: a 5-12-13 triangle. For $\begin{bmatrix} 5 \\ 12 \end{bmatrix}$: the squares are 25 and 144, their sum is 169, and $\sqrt{169} = 13$. Like 3-4-5, this is one of the few right triangles whose sides are all whole numbers.

Worked example: a vector in $\mathbb{R}^3$ with a negative component. For $\begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$: the squares are $(-2)^2 = 4$, $1^2 = 1$, and $2^2 = 4$; their sum is 9, and $\sqrt{9} = 3$. The minus sign disappears when we square, as it should: a length cannot depend on which way along an axis the arrow happens to point.

Worked example: the climbing plane. The plane that climbs at 10 m/s while flying 150 m/s east and 200 m/s north has velocity $\begin{bmatrix} 150 \\ 200 \\ 10 \end{bmatrix}$, so its speed is

$$\sqrt{150^2 + 200^2 + 10^2} = \sqrt{22500 + 40000 + 100} = \sqrt{62600} \approx 250.2 \text{ m/s.}$$

Climbing adds only about 0.2 m/s to the speed. Squaring makes a small component matter even less: $10^2 = 100$ is tiny next to 62500. This is worth remembering for data: a component that is small compared to the others barely affects a length.

Worked example: four equal components. For $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ in $\mathbb{R}^4$: $1 + 1 + 1 + 1 = 4$, and $\sqrt{4} = 2$. The formula does not care that we cannot draw $\mathbb{R}^4$.

Watch out: a length is not a sum

The length of $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ is 5, not $3 + 4 = 7$: the diagonal of a rectangle is shorter than going along two sides. Likewise $\|\begin{bmatrix} -2 \\ 1 \end{bmatrix}\| = 3$, which is neither $-2 + 1 + 2 = 1$ nor $2 + 1 + 2 = 5$. And the square root goes on the **total**, once: $\sqrt{3^2 + 4^2} = 5$, whereas $\sqrt{3^2} + \sqrt{4^2} = 7$ is the same wrong

answer in disguise.

Notebook U2-2_Vectors-1_ArrowsAndOps, §5, magnitude as a hypotenuse

Exercise 3.1

Find the magnitude of each vector.

1. $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$ 2. $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ 3. $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ 4. $\begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}$ 5. the email $\bar{d}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 2 \end{bmatrix}$

Solution to Exercise 3.1

Square, add, take the root.

1. $\sqrt{36 + 64} = \sqrt{100} = 10$. (It is $2 \times \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, and its length is 2×5 .)
2. $\sqrt{1 + 1} = \sqrt{2} \approx 1.414$.
3. $\sqrt{0.36 + 0.64} = \sqrt{1} = 1$. A vector of length exactly 1: Section 3.6 calls it a unit vector.
4. $\sqrt{4 + 9 + 36} = \sqrt{49} = 7$.
5. $\sqrt{4 + 0 + 1 + 4} = \sqrt{9} = 3$, the same length as $\bar{d}_1$ although the words are different.

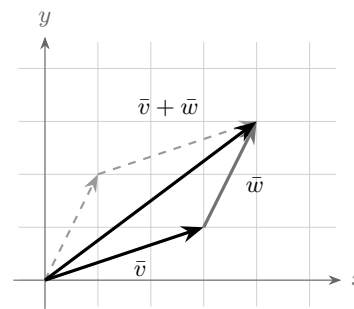
3.2 Addition: Head to Tail

Vectors add **componentwise**, as you would guess: first components together, second components together.

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3+1 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Call the first vector $\bar{v}$ and the second $\bar{w}$ (in this picture only; they are not the plane). The picture of addition uses the freedom to slide arrows:

- Put the tail of $\bar{w}$ on the head of $\bar{v}$.
- The sum is the arrow from the first tail to the last head.
- Either order works: $\bar{v} + \bar{w} = \bar{w} + \bar{v}$. The dashed route takes $\bar{w}$ first and lands in the same place.



Either route reaches (4,3).

In the running examples, addition means combining:

- **Velocity**: a plane's velocity through the **air**, $\bar{v}$, plus the velocity of the wind, $\bar{w}$, is its velocity over the **ground**, $\bar{v} + \bar{w}$. The plane moves through the air while the air itself carries it along, and the two motions add. The worked example below puts numbers on it.

- **Email:** paste $\bar{d}_1$ and $\bar{d}_3$ together into one email and the counts add: 4 *free*, 1 *win*, 3 *cash*, 2 *meeting*. In symbols,

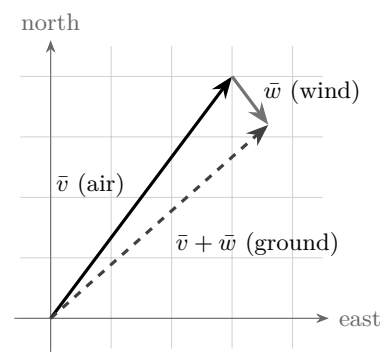
$$\bar{d}_1 + \bar{d}_3 = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 3 \\ 2 \end{bmatrix}.$$

Worked example: the plane in a wind. The plane's velocity through the air is $\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$. A wind blows toward the east at 30 m/s and toward the south at 40 m/s, so $\bar{w} = \begin{bmatrix} 30 \\ -40 \end{bmatrix}$ (south is negative north). The velocity over the ground is

$$\bar{v} + \bar{w} = \begin{bmatrix} 150 + 30 \\ 200 + (-40) \end{bmatrix} = \begin{bmatrix} 180 \\ 160 \end{bmatrix}.$$

The ground speed is its length,

$$\sqrt{180^2 + 160^2} = \sqrt{58000} \approx 240.8 \text{ m/s}.$$



The wind itself has speed $\sqrt{30^2 + 40^2} = 50$ m/s, and it pushes partly east, the direction the plane is partly going. Yet the ground speed *dropped*, from 250 to about 241 m/s: overall this wind works against the plane more than with it. Section 4.3 gives a single number, the dot product, that detects this in advance.

One caution about words. We describe a wind by the direction it blows *toward*, because that is the direction of its velocity vector. Weather reports do the opposite and name a wind by where it comes *from*: a “north wind” blows toward the south. When a problem says “a wind from the north,” its vector points south.

Worked example: adding meals. Nutrition labels are vectors. Record each meal as grams of (protein, carbohydrate, fat). Breakfast is $\begin{bmatrix} 20 \\ 45 \\ 10 \end{bmatrix}$ and lunch is $\begin{bmatrix} 30 \\ 60 \\ 15 \end{bmatrix}$, so the day so far is

$$\begin{bmatrix} 20 \\ 45 \\ 10 \end{bmatrix} + \begin{bmatrix} 30 \\ 60 \\ 15 \end{bmatrix} = \begin{bmatrix} 50 \\ 105 \\ 25 \end{bmatrix} \quad \begin{array}{l} \leftarrow \text{protein (g)} \\ \leftarrow \text{carbohydrate (g)} \\ \leftarrow \text{fat (g)} \end{array}$$

Every nutrient is totalled at once, and protein is only ever added to protein. That is what “componentwise” buys you: the k th entry of the sum depends only on the k th entries of the vectors being added. (Section 4.1 turns these grams into calories.)

Worked example: signs in $\mathbb{R}^3$. Negative components need care, and nothing more:

$$\begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + \begin{bmatrix} -5 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 + (-5) \\ -1 + 3 \\ 4 + 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 5 \end{bmatrix}.$$

The picture also shows a basic fact about lengths. The arrows $\bar{v}$, $\bar{w}$, and $\bar{v} + \bar{w}$ form a triangle, and one side of a triangle is never longer than the other two sides put together.

The triangle inequality

$$\|\bar{v} + \bar{w}\| \leq \|\bar{v}\| + \|\bar{w}\|$$

In the head-to-tail picture, $\|\bar{v} + \bar{w}\| = \sqrt{4^2 + 3^2} = 5$, while going along $\bar{v}$ and then $\bar{w}$ covers $\sqrt{10} + \sqrt{5} \approx 3.16 + 2.24 \approx 5.4$. The direct route wins. For the plane, the ground speed 240.8 is well under $250 + 50 = 300$: the air speed and the wind speed simply add only if the wind blows exactly along the plane's heading. In general the two sides are equal only when $\bar{v}$ and $\bar{w}$ point the same way; for example $\begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 9 \\ 12 \end{bmatrix}$ has length $15 = 5 + 10$.

Watch out: plain Python lists do not add like vectors

In plain Python, `[1, 2] + [3, 4]` gives `[1, 2, 3, 4]`: the `+` sticks the two lists end to end. To add componentwise, use numpy arrays: `np.array([1, 2]) + np.array([3, 4])` gives `[4 6]`. numpy goes the other way in one respect: `np.array([150, 200]) + 1` adds 1 to every entry and returns `[151 201]`, although “a vector plus a number” is not defined in mathematics. numpy is being convenient (this is called broadcasting), not doing vector algebra.

Notebook U2-2_Vectors-1_ArrowsAndOps, §2, addition drawn head to tail

Exercise 3.2

1. Compute $\begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 5 \end{bmatrix}$, and check that adding in the other order gives the same answer.
2. A plane flies due east through the air at 200 m/s, and a wind blows toward the south at 50 m/s. Find the plane's velocity over the ground and its ground speed.

Solution to Exercise 3.2

1. $\begin{bmatrix} 4+(-1) \\ -2+5 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, and $\begin{bmatrix} -1+4 \\ 5+(-2) \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ too. Order cannot matter, since each entry is an ordinary sum of two numbers, and $a + b = b + a$ for numbers.
2. Air velocity $\begin{bmatrix} 200 \\ 0 \end{bmatrix}$, wind $\begin{bmatrix} 0 \\ -50 \end{bmatrix}$, so the ground velocity is $\begin{bmatrix} 200 \\ -50 \end{bmatrix}$: still 200 m/s east, and now drifting 50 m/s south. The ground speed is $\sqrt{200^2 + 50^2} = \sqrt{42500} \approx 206.2$ m/s. The wind blows at right angles to the heading, so it barely changes the speed (by about 6 m/s) but pushes the plane off course; Section 4.3 calls this a crosswind.

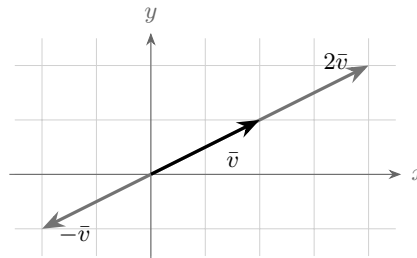
3.3 Scalar Multiplication: Stretch, Shrink, Flip

A **scalar** is a plain number, as opposed to a vector. Multiplying a vector by a scalar c multiplies every component by c , which **scales** the arrow; that is where the name comes from.

$$c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c v_1 \\ c v_2 \end{bmatrix}$$

What happens to the arrow depends on c :

- $c > 1$ stretches it; $0 < c < 1$ shrinks it.
- $c < 0$ flips it to point the opposite way; $c = 0$ gives $\vec{0}$.
- Every multiple stays on **one line through the origin**. Remember this for **span**, in Section 5.2.



All multiples of $\vec{v}$ lie on one line.

How much does the length change? Every component is multiplied by c , so every square is multiplied by c^2 , and taking the square root brings out $\sqrt{c^2} = |c|$, the size of c without its sign:

$$\|c\vec{v}\| = |c| \|\vec{v}\|.$$

For example, $-2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -6 \\ -8 \end{bmatrix}$ has length $\sqrt{36 + 64} = 10 = 2 \times 5$. Flipping never makes a length negative.

In the running examples:

- **Velocity:** $2\vec{v}$ doubles the speed on the same heading; $-\vec{v}$ reverses course. Half speed on the same heading is $\frac{1}{2}\vec{v} = \begin{bmatrix} 75 \\ 100 \end{bmatrix}$, which has length 125 m/s.
- **Units:** converting m/s to km/h is scaling by 3.6, because 1 m/s is 3600 m per hour, or 3.6 km/h. So 250 m/s is $3.6 \times 250 = 900$ km/h, and the plane's velocity becomes

$$3.6 \begin{bmatrix} 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 540 \\ 720 \end{bmatrix} \text{ km/h,}$$

the same motion in different units. Check: $\sqrt{540^2 + 720^2} = 900$.

- **Email:** $\vec{d}_2 = 2\vec{d}_1$, since every word appears twice as often. Same **direction**, twice the length, which matches $\|\vec{d}_2\| = 6 = 2 \times 3$ from Section 3.1.

Worked example: a weight-based dose. Many drugs are dosed per kilogram of body weight. A three-drug regimen calls for 5 mg/kg of the first drug, 0.5 mg/kg of the second, and 10 mg/kg of the third. For a 70 kg patient, scale the per-kilogram vector by 70:

$$70 \begin{bmatrix} 5 \\ 0.5 \\ 10 \end{bmatrix} = \begin{bmatrix} 350 \\ 35 \\ 700 \end{bmatrix} \text{ mg.}$$

Every drug is scaled by the same factor, so the *proportions* between the three doses (1 to 0.1 to 2) are exactly the same for every patient; only the overall size changes. That is scalar multiplication in one sentence: same direction, new length.

Watch out: $2 * [150, 200]$ in plain Python

Multiplying a plain Python list by a whole number repeats the list: $2 * [150, 200]$ is $[150, 200, 150, 200]$, a list of four numbers. To scale a vector, make it a numpy array first: $2 * \text{np.array}([150, 200])$ is $[300 \ 400]$.

Exercise 3.3

1. Compute $-3 \begin{bmatrix} 2 \\ -1 \\ 0.5 \end{bmatrix}$.
2. A velocity is recorded as $\begin{bmatrix} 720 \\ -360 \end{bmatrix}$ km/h. Convert it to m/s.
3. Is $\begin{bmatrix} -4 \\ -6 \end{bmatrix}$ a scalar multiple of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$? Is $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$?

Solution to Exercise 3.3

1. Multiply every component by -3 : $\begin{bmatrix} -6 \\ 3 \\ -1.5 \end{bmatrix}$. Each sign flips, because c is negative.
2. Going from m/s to km/h multiplies by 3.6, so going back divides by 3.6, that is, scales by $1/3.6$: $\begin{bmatrix} 720/3.6 \\ -360/3.6 \end{bmatrix} = \begin{bmatrix} 200 \\ -100 \end{bmatrix}$ m/s.
3. $\begin{bmatrix} -4 \\ -6 \end{bmatrix} = -2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, so **yes**, with $c = -2$ (same line, opposite way). For $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ we would need $2c = 4$ and $3c = 5$ at once; the first gives $c = 2$ and the second $c = 5/3$, a contradiction, so **no**. The test is always the same: one value of c must work for every component.

3.4 Subtraction: Add the Reverse

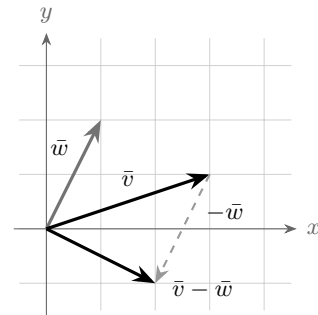
Subtraction is also componentwise:

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 - 1 \\ 1 - 2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

To draw it we do not need anything new, because subtracting is adding the reverse:

$$\bar{v} - \bar{w} = \bar{v} + (-\bar{w}).$$

Here $-\bar{w}$ is $\bar{w}$ scaled by -1 : the same length, pointing the opposite way. Put its tail on the head of $\bar{v}$ and add head to tail, exactly as before.



$\bar{v}$, then the reversed $\bar{w}$ (dashed).

Two examples show what a difference means.

Velocity: a plane in a headwind. A plane flies 250 m/s east, and a 50 m/s **headwind** (a wind blowing straight at it) takes some of that away:

$$\begin{bmatrix} 250 \\ 0 \end{bmatrix} - \begin{bmatrix} 50 \\ 0 \end{bmatrix} = \begin{bmatrix} 200 \\ 0 \end{bmatrix}$$

The result is the plane's velocity over the **ground**.

Patient: what changed. Let $\vec{p}'$ be the same patient measured a year later. Subtracting today's vector from next year's gives the change in every measurement at once:

$$\vec{p}' - \vec{p} = \begin{bmatrix} 55 \\ 136 \\ 6.5 \end{bmatrix} - \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 0.4 \end{bmatrix}$$

The patient is up 1 year, 6 mmHg, and 0.4 in A1c.

Worked example: recovering the wind. Subtraction undoes addition. A plane's instruments give its velocity through the air, $\begin{bmatrix} 150 \\ 200 \end{bmatrix}$, and GPS gives its velocity over the ground, $\begin{bmatrix} 180 \\ 160 \end{bmatrix}$. The wind is whatever was added to the first to produce the second:

$$\text{wind} = \text{ground} - \text{air} = \begin{bmatrix} 180 \\ 160 \end{bmatrix} - \begin{bmatrix} 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 30 \\ -40 \end{bmatrix},$$

exactly the wind of Section 3.2. Writing $\vec{v}$ for the air velocity, $\vec{w}$ for the wind, and $\vec{g}$ for the ground velocity, addition said $\vec{v} + \vec{w} = \vec{g}$, and subtracting $\vec{v}$ from both sides gives $\vec{w} = \vec{g} - \vec{v}$, just as with ordinary numbers.

Worked example: a lifestyle program. A patient's (weight in kg, systolic BP in mmHg, LDL cholesterol in mg/dL) is $\begin{bmatrix} 98 \\ 148 \\ 160 \end{bmatrix}$ at enrollment and $\begin{bmatrix} 91 \\ 136 \\ 128 \end{bmatrix}$ six months later. The change is after minus before:

$$\begin{bmatrix} 91 \\ 136 \\ 128 \end{bmatrix} - \begin{bmatrix} 98 \\ 148 \\ 160 \end{bmatrix} = \begin{bmatrix} -7 \\ -12 \\ -32 \end{bmatrix}.$$

Every component is negative, so every measurement went down: 7 kg lost, blood pressure down 12 mmHg, LDL down 32 mg/dL. One subtraction summarizes the whole visit.

Watch out: order matters

Unlike addition, subtraction is not symmetric: $\vec{w} - \vec{v} = -(\vec{v} - \vec{w})$, the same arrow pointing the other way. In the lifestyle example, before minus after gives $\begin{bmatrix} 7 \\ 12 \\ 32 \end{bmatrix}$, which reads as if everything *rose*. Pick a convention and say it out loud; “after minus before” (or “end minus start”) is the usual one, and it matches the walking rule of the point-versus-vector subsection.

3.5 Distance Between Two Points

With subtraction in hand we can measure how far apart two points are. Think of $\vec{v}$ and $\vec{w}$ as the points at their heads. Since $\vec{w} + (\vec{v} - \vec{w}) = \vec{v}$, the vector $\vec{v} - \vec{w}$ is the trip from the head of $\vec{w}$ to the head of $\vec{v}$, and the distance is the length of that trip.

Definition: Distance

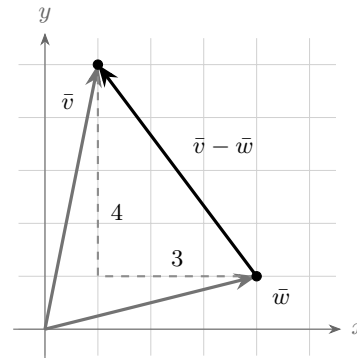
The **distance** between two points is the length of the trip between them:

$$d(\vec{v}, \vec{w}) = \|\vec{v} - \vec{w}\| = \sqrt{\sum_i (v_i - w_i)^2}$$

The symbol $\sum_i$ means “add up over every component i ,” so in the plane the formula reads

$$d(\bar{v}, \bar{w}) = \sqrt{(v_1 - w_1)^2 + (v_2 - w_2)^2} :$$

subtract matching components, square, add, and take the square root. This is **Euclidean distance**, ordinary straight-line distance, and it is what k -means and nearest-neighbor methods measure. Because each difference is squared, the order does not matter: $d(\bar{v}, \bar{w}) = d(\bar{w}, \bar{v})$.



Legs 3 and 4: the points are 5 apart.

Running example: $\bar{p}$ and a second patient $\bar{q}$, and a warning. Take a second patient $\bar{q}$ (patient B of Section 1), aged 58 with blood pressure 138 and A1c 6.4:

$$\bar{p} - \bar{q} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} - \begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix} = \begin{bmatrix} -4 \\ -8 \\ -0.3 \end{bmatrix}, \quad d(\bar{p}, \bar{q}) = \sqrt{16 + 64 + 0.09} \approx 8.95$$

Look at where that total comes from:

- Blood pressure supplies 64 of the roughly 80 under the square root; A1c supplies 0.09 and barely registers.
- That is not because blood pressure matters more. It is only because blood pressure is recorded in bigger numbers. The biggest units win, so **standardize** (put every measurement on a comparable scale) before measuring distance.

Worked example: distance between two points in the plane. From (1, 2) to (4, 6): the difference is $\begin{bmatrix} 1-4 \\ 2-6 \end{bmatrix} = \begin{bmatrix} -3 \\ -4 \end{bmatrix}$, so the distance is $\sqrt{9 + 16} = 5$. Subtracting the other way gives $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and the same 5.

Worked example: distance in $\mathbb{R}^3$. Between (1, 0, 2) and (3, -1, 4): the difference is $\begin{bmatrix} -2 \\ 1 \\ -2 \end{bmatrix}$, the squares are 4, 1, 4, and the distance is $\sqrt{9} = 3$.

Worked example: standardizing, step by step. One simple way to put measurements on a comparable scale is to divide each one by how much it *typically varies* among patients: a difference of one “typical variation” then counts the same for every measurement. Suppose that in this clinic ages typically vary by about 10 years, systolic blood pressure by about 16 mmHg, and A1c by about 0.6 percentage points. (Unit 4 makes “typically varies” precise as the *standard deviation*. The usual recipe also subtracts an average from every measurement first; that shifts both patients by the same amount, so it cannot change the distance between them.) Divide each component of $\bar{p} - \bar{q}$ by its typical variation:

$$\begin{bmatrix} -4/10 \\ -8/16 \\ -0.3/0.6 \end{bmatrix} = \begin{bmatrix} -0.4 \\ -0.5 \\ -0.5 \end{bmatrix}, \quad \text{standardized distance} = \sqrt{0.16 + 0.25 + 0.25} = \sqrt{0.66} \approx 0.81.$$

The share each measurement contributes is now sensible:

	Age	BP	A1c	Total under the root
Raw squared differences	16	64	0.09	80.09
Standardized squared differences	0.16	0.25	0.25	0.66

Raw, A1c contributes about 0.1% of the total. Standardized, it contributes more than a third, on an equal footing with blood pressure.

Worked example: standardizing changes who is nearest. Now the payoff. A nearest-neighbor method must decide which of two patients is more like $\bar{p} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix}$:

$$\bar{r} = \begin{bmatrix} 54 \\ 139 \\ 6.1 \end{bmatrix} \text{ (same age and A1c, BP 9 higher),} \quad \bar{s} = \begin{bmatrix} 55 \\ 130 \\ 8.1 \end{bmatrix} \text{ (a year older, A1c 2 points higher).}$$

- **Raw:** $d(\bar{p}, \bar{r}) = \sqrt{0 + 81 + 0} = 9$ and $d(\bar{p}, \bar{s}) = \sqrt{1 + 0 + 4} = \sqrt{5} \approx 2.24$. Raw distance calls $\bar{s}$ far more similar.
- **Standardized** (each difference divided by 10, 16, and 0.6 as before):

$$d(\bar{p}, \bar{r}) = \sqrt{0 + \left(\frac{9}{16}\right)^2 + 0} = 0.5625,$$

$$d(\bar{p}, \bar{s}) = \sqrt{\left(\frac{1}{10}\right)^2 + 0 + \left(\frac{2}{0.6}\right)^2} \approx \sqrt{0.01 + 11.11} \approx 3.33.$$

Now $\bar{r}$ is much closer.

Clinically, the standardized answer is the sensible one. An A1c of 6.1 is in the prediabetes range, while 8.1 is well above the diabetes threshold of 6.5: patient $\bar{s}$ has a different condition, while $\bar{r}$ differs by an unremarkable 9 mmHg. The raw distance missed this only because A1c is written in small numbers.

Watch out: distance is not a difference of lengths

The distance between two points is the length of their *difference*, $\|\bar{v} - \bar{w}\|$, not the difference of their lengths, $\|\bar{v}\| - \|\bar{w}\|$. The points $(3, 4)$ and $(4, 3)$ both sit 5 from the origin, so their lengths differ by 0, yet they are $\sqrt{1 + 1} = \sqrt{2} \approx 1.41$ apart. Two points at the same distance from the origin can be anywhere on a circle.

Exercise 3.4

1. Find the distance between the points $(2, -1)$ and $(-1, 3)$.
2. Find the distance between the email vectors $\bar{d}_1$ and $\bar{e} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ from Exercise 2.2.
3. Two blood-pressure readings, recorded as (systolic, diastolic), are $\begin{bmatrix} 142 \\ 92 \end{bmatrix}$ in the morning and $\begin{bmatrix} 128 \\ 84 \end{bmatrix}$ in the evening. Find the change from morning to evening and the distance between the readings. Is the raw distance meaningful here, or must we standardize first?

Solution to Exercise 3.4

1. The difference is $\begin{bmatrix} 2-(-1) \\ -1-3 \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, so the distance is $\sqrt{9+16} = 5$.
2. $\bar{d}_1 - \bar{e} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$, so the distance is $\sqrt{0+1+1+1} = \sqrt{3} \approx 1.73$.
3. Evening minus morning is $\begin{bmatrix} -14 \\ -8 \end{bmatrix}$: systolic down 14 mmHg, diastolic down 8. The distance is $\sqrt{196+64} = \sqrt{260} \approx 16.1$ mmHg. Here the raw distance **is** meaningful: both components are pressures in the same unit and of similar size, so neither swamps the other just by how it is written. Standardizing matters when components are in different units or on very different scales, as with age, blood pressure, and A1c.

3.6 Unit Vectors: Getting $\hat{v}$ from $\bar{v}$

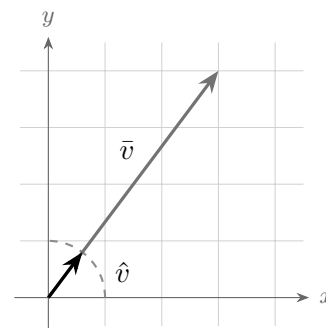
Sometimes only the direction of a vector matters, and its size just gets in the way. A **unit vector** is a vector of length exactly 1, and the **hat** marks one: $\hat{v}$ keeps the **direction** of $\bar{v}$ and drops its size.

Normalizing: divide by the length

$$\hat{v} = \frac{1}{\|\bar{v}\|} \bar{v} = \frac{\bar{v}}{\|\bar{v}\|}$$

This is ordinary **scaling**, by the positive number $1/\|\bar{v}\|$. Scaling by a positive number multiplies every component, and therefore the length, by that number (Section 3.3). So the direction stays the same and the length becomes $\|\bar{v}\|/\|\bar{v}\| = 1$.

Normalizing is **not defined for** $\bar{0}$: its length is 0, we cannot divide by 0, and the zero vector has no direction to keep anyway.



$\hat{v}$ ends on the arc of radius 1.

In the running examples:

- **Velocity**: the plane's **heading** is

$$\hat{v} = \frac{1}{250} \begin{bmatrix} 150 \\ 200 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix},$$

which says which way the plane is going with the speed stripped out. Check the length: $0.6^2 + 0.8^2 = 0.36 + 0.64 = 1$. (In the picture, drawn at one square per 50 m/s, $\bar{v}$ reaches (3, 4) and $\hat{v}$ reaches (0.6, 0.8).)

- **Email**: $\bar{d}_2 = 2\bar{d}_1$, so $\hat{d}_2 = \hat{d}_1$. Indeed $\hat{d}_1 = \frac{1}{3}\bar{d}_1$ and $\hat{d}_2 = \frac{1}{6}\bar{d}_2 = \frac{1}{6}(2\bar{d}_1) = \frac{1}{3}\bar{d}_1$. Normalizing keeps the same mix of words and forgets the length.

Notice that the heading has **no units**: m/s divided by m/s cancels. A unit vector is a pure direction. Turning this around gives a useful way to think of any nonzero vector:

$$\bar{v} = \|\bar{v}\| \hat{v} \quad (\text{length times direction}), \quad \text{for the plane: } 250 \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}.$$

Worked example: same heading, new speed. Air traffic control asks the plane to hold its heading but speed up to 300 m/s. Length times direction gives the new velocity directly:

$$300 \hat{v} = 300 \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 180 \\ 240 \end{bmatrix}, \quad \text{check: } \sqrt{180^2 + 240^2} = 300.$$

Separating a vector into its length and its direction lets us change one without disturbing the other.

Worked example: normalizing a 5-12-13 vector. Let $\bar{u} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$. Its length is 13 (Section 3.1), so

$$\hat{u} = \frac{1}{13} \begin{bmatrix} 5 \\ 12 \end{bmatrix} = \begin{bmatrix} 5/13 \\ 12/13 \end{bmatrix} \approx \begin{bmatrix} 0.385 \\ 0.923 \end{bmatrix}, \quad \text{check: } \frac{25}{169} + \frac{144}{169} = \frac{169}{169} = 1.$$

Keeping the fractions makes the check exact; the decimals are for reading.

Worked example: normalizing in $\mathbb{R}^3$ and $\mathbb{R}^4$. The vector $\begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix}$ has length $\sqrt{9+0+16} = 5$, and the email $\bar{d}_1$ has length 3, so

$$\frac{1}{5} \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix} = \begin{bmatrix} -0.6 \\ 0 \\ 0.8 \end{bmatrix}, \quad \hat{d}_1 = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \\ 0 \end{bmatrix} \approx \begin{bmatrix} 0.667 \\ 0.333 \\ 0.667 \\ 0 \end{bmatrix}.$$

Check the second: $\frac{4}{9} + \frac{1}{9} + \frac{4}{9} + 0 = 1$.

Watch out: “normalize” means length 1, not sum 1

The heading $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ has components summing to 1.4, not 1, and that is correct: a unit vector has *length* 1. Dividing by the *sum* of the components instead, $\frac{1}{350} \begin{bmatrix} 150 \\ 200 \end{bmatrix} \approx \begin{bmatrix} 0.429 \\ 0.571 \end{bmatrix}$, gives proportions that add to 1 (handy for other purposes), but that vector has length about 0.714, so it is not a unit vector. Always divide by the length, $\sqrt{\text{sum of squares}}$.

Exercise 3.5

1. Normalize $\begin{bmatrix} 8 \\ -6 \end{bmatrix}$.
2. A plane on heading $\begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix}$ flies at 200 m/s. What is its velocity?
3. Is $\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$ a unit vector?

Solution to Exercise 3.5

1. The length is $\sqrt{64+36} = 10$, so the unit vector is $\begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix}$. Check: $0.64 + 0.36 = 1$.
2. Length times direction: $200 \begin{bmatrix} 0.6 \\ -0.8 \end{bmatrix} = \begin{bmatrix} 120 \\ -160 \end{bmatrix}$ m/s, that is 120 m/s east and 160 m/s south.

3. **No.** Its length is $\sqrt{0.25 + 0.25} = \sqrt{0.5} \approx 0.707$. Its components add to 1, which is not the test (see the Watch out box). The unit vector in its direction is $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$.

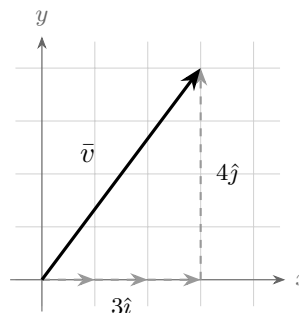
3.7 The Standard Unit Vectors

Two unit vectors are special enough to have their own names: one step along each axis.

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

They are read “i-hat” and “j-hat” (the dot on the letter is dropped to make room for the hat). By scaling and adding them we can build **every** vector in the plane:

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3\hat{i} + 4\hat{j}$$



Three steps of $\hat{i}$, then four of $\hat{j}$.

So the components of a vector are really **amounts**: how much of $\hat{i}$ and how much of $\hat{j}$. That is a **choice of axes**, and in Section 6 we will change it.

In the running examples:

- **Velocity:** $150\hat{i} + 200\hat{j}$ reads “150 m/s east plus 200 m/s north.”
- **Email:** each word gets its own unit vector: the vector of an email that says just that word, once. The counts are the amounts. For $\bar{d}_1$,

$$\bar{d}_1 = 2 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\text{free}} + 1 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}}_{\text{win}} + 2 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\text{cash}} + 0 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}}_{\text{meeting}} = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix}.$$

Worked example: translating both ways. From column to steps: $\begin{bmatrix} -2 \\ 5 \end{bmatrix} = -2\hat{i} + 5\hat{j}$, two steps in the negative x direction and five up. From steps to column: $7\hat{i} - 3\hat{j} = \begin{bmatrix} 7 \\ -3 \end{bmatrix}$. A component of 0 simply means “no steps that way”: $\begin{bmatrix} 0 \\ -200 \end{bmatrix} = 0\hat{i} - 200\hat{j}$ is the plane flying due south.

Worked example: the patient, one axis per measurement. The same idea works in $\mathbb{R}^3$ with three one-step vectors, one per measurement:

$$\bar{p} = 54 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 130 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 6.1 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix}.$$

Each one-step vector stands for “one unit of this measurement and nothing else,” and the patient’s values are the amounts.

Notebook U2-2_Vectors-1_ArrowsAndOps, §4, building $\bar{v}$ from $\hat{i}$ and $\hat{j}$

3.8 Exercise: Vector Arithmetic

Exercise 3.6

Let

$$\bar{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{and} \quad \bar{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

1. Compute $\bar{u} + \bar{v}$ and $\bar{u} - \bar{v}$.
2. Compute $2\bar{u} - 3\bar{v}$.
3. Compute $\|\bar{u}\|$.
4. Find the distance between the points at the heads of $\bar{u}$ and $\bar{v}$.
5. Write $\bar{u}$ in terms of $\hat{i}$ and $\hat{j}$.
6. What goes wrong if you try to add $\bar{u}$ to $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$?

Solution to Exercise 3.6

1. Add and subtract component by component. Watch the sign in the second entry of the difference: subtracting -1 is the same as adding 1.

$$\bar{u} + \bar{v} = \begin{bmatrix} 1 + 4 \\ 3 + (-1) \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}, \quad \bar{u} - \bar{v} = \begin{bmatrix} 1 - 4 \\ 3 - (-1) \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}.$$

2. Scale first, then subtract:

$$2\bar{u} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}, \quad 3\bar{v} = \begin{bmatrix} 12 \\ -3 \end{bmatrix}, \quad \text{so} \quad 2\bar{u} - 3\bar{v} = \begin{bmatrix} 2 - 12 \\ 6 - (-3) \end{bmatrix} = \begin{bmatrix} -10 \\ 9 \end{bmatrix}.$$

3. Square the components, add, and take the square root: $\|\bar{u}\| = \sqrt{1^2 + 3^2} = \sqrt{10} \approx 3.16$. Leave it exact, as $\sqrt{10}$, unless a decimal is asked for.
4. The distance between two points is the length of the difference of their vectors, and part 1 already found $\bar{u} - \bar{v}$:

$$d(\bar{u}, \bar{v}) = \|\bar{u} - \bar{v}\| = \sqrt{(-3)^2 + 4^2} = \sqrt{25} = 5.$$

A 3-4-5 triangle again. Subtracting in the other order gives $\bar{v} - \bar{u} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$, which has the same length, so the distance does not depend on which point you start from.

5. $\bar{u} = 1\hat{i} + 3\hat{j}$: one step along the x -axis and three along the y -axis. The components are the amounts.
6. **Nothing is defined.** $\bar{u}$ lives in $\mathbb{R}^2$ and the other vector in $\mathbb{R}^3$, so there is no third component of $\bar{u}$ to add to the 3. Vector operations need **matching dimensions**. In **numpy**, trying it raises a broadcast error (“operands could not be broadcast together with shapes (2,) (3,)”), which is numpy’s way of saying the two shapes do not fit together.

3.9 More Practice: Operations

Exercise 3.7

A plane's velocity through the air is $\bar{a} = \begin{bmatrix} 120 \\ 160 \end{bmatrix}$ m/s, and the wind is $\bar{w} = \begin{bmatrix} -20 \\ -10 \end{bmatrix}$ m/s.

1. Find the velocity over the ground and the ground speed.
2. Find the airspeed $\|\bar{a}\|$ and the heading $\hat{a}$.
3. How much speed did the wind cost the plane?

Solution to Exercise 3.7

1. Air plus wind: $\begin{bmatrix} 120+(-20) \\ 160+(-10) \end{bmatrix} = \begin{bmatrix} 100 \\ 150 \end{bmatrix}$. The ground speed is $\sqrt{100^2 + 150^2} = \sqrt{32500} \approx 180.3$ m/s.
2. $\|\bar{a}\| = \sqrt{14400 + 25600} = \sqrt{40000} = 200$ m/s, and $\hat{a} = \frac{1}{200} \begin{bmatrix} 120 \\ 160 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, the same heading as our running plane.
3. $200 - 180.3 \approx 19.7$ m/s. This wind blows mostly toward the west and a little toward the south, broadly against a plane heading northeast, so it slows the plane. (Note that this is a difference of two *speeds*, which is a fair question to ask about speeds; it is not a distance between vectors.)

Exercise 3.8

Morning and evening blood-pressure readings (systolic, diastolic) are $\bar{m} = \begin{bmatrix} 142 \\ 92 \end{bmatrix}$ and $\bar{e} = \begin{bmatrix} 128 \\ 84 \end{bmatrix}$.

1. Compute $\frac{1}{2}\bar{m} + \frac{1}{2}\bar{e}$. What does it represent?
2. Find a number $c > 0$ with $\|c \begin{bmatrix} 3 \\ 4 \end{bmatrix}\| = 1$, and another with $\|c \begin{bmatrix} 3 \\ 4 \end{bmatrix}\| = 10$.

Solution to Exercise 3.8

1. Scale each by $\frac{1}{2}$ and add: $\begin{bmatrix} 71 \\ 46 \end{bmatrix} + \begin{bmatrix} 64 \\ 42 \end{bmatrix} = \begin{bmatrix} 135 \\ 88 \end{bmatrix}$. It is the **average** reading, 135/88: each component is the average of the two readings of that component. Section 5 calls a scale-and-add like this a linear combination.
2. Since $\|c\bar{v}\| = |c| \|\bar{v}\|$ and $\|\begin{bmatrix} 3 \\ 4 \end{bmatrix}\| = 5$, we need $5c = 1$, so $c = 1/5 = 0.2$ (this gives the unit vector $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$); and $5c = 10$, so $c = 2$ (giving $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$).

4 The Dot Product and the Angle

4.1 One Number That Compares Two Vectors

Every operation so far has taken vectors in and given a vector back. The dot product is different: it takes two vectors and returns a single **number**, and that number measures how much the two vectors agree.

Definition: Dot product

For $\vec{v}, \vec{w} \in \mathbb{R}^n$, the **dot product** (or **inner product**) is

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n = \sum_{i=1}^n v_i w_i$$

How to read it:

- Multiply matching components, then add.
- The result is a **number**, not a vector.
- Both vectors need the **same** number of components; otherwise some components have nothing to be matched with.

Running example: two emails. Multiply the counts word by word and add:

$$\vec{d}_1 \cdot \vec{d}_3 = \underbrace{(2)(2)}_{\text{free}} + \underbrace{(1)(0)}_{\text{win}} + \underbrace{(2)(1)}_{\text{cash}} + \underbrace{(0)(2)}_{\text{meeting}} = 6$$

Each word adds (its count in one email) $\times$ (its count in the other). A word therefore contributes only if it appears in *both* emails: here *free* and *cash* contribute, while *win* and *meeting* each appear in only one email and add nothing. It follows that two emails with **no words in common** score 0.

Worked example: small numeric dot products. In the plane, $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = (1)(3) + (2)(4) = 3 + 8 = 11$. In $\mathbb{R}^3$, signs included,

$$\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 1 \end{bmatrix} = (2)(4) + (-1)(5) + (3)(1) = 8 - 5 + 3 = 6.$$

A negative product pulls the total down: matching components that disagree in sign count *against* agreement.

Worked example: a pharmacy bill. A dot product is a **weighted total**: one vector says how many, the other says what each one is worth. A pharmacy order is 30, 60, and 90 tablets of three drugs costing \$0.50, \$0.20, and \$1.10 per tablet:

$$\underbrace{\begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}}_{\text{quantities}} \cdot \underbrace{\begin{bmatrix} 0.50 \\ 0.20 \\ 1.10 \end{bmatrix}}_{\text{price each}} = 15 + 12 + 99 = \$126.$$

Each drug contributes (how many) $\times$ (price each), and the dot product adds up the bill.

Worked example: calories from a day of meals. In Section 3.2 breakfast plus lunch came to $\begin{bmatrix} 50 \\ 105 \\ 25 \end{bmatrix}$ grams of (protein, carbohydrate, fat). Protein and carbohydrate supply about 4 kcal per gram and fat about 9, so the calories are

$$\begin{bmatrix} 50 \\ 105 \\ 25 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 4 \\ 9 \end{bmatrix} = 200 + 420 + 225 = 845 \text{ kcal.}$$

Worked example: a risk score. Many prediction models compute exactly this kind of weighted total: a vector of **weights**, one per measurement, dotted with a patient’s vector. With made-up weights $\begin{bmatrix} 0.02 \\ 0.01 \\ 0.3 \end{bmatrix}$ for (age, BP, A1c), patient $\bar{p}$ scores

$$\begin{bmatrix} 0.02 \\ 0.01 \\ 0.3 \end{bmatrix} \cdot \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} = 1.08 + 1.30 + 1.83 = 4.21.$$

Unlike the length of $\bar{p}$, this mixes units harmlessly: each weight carries whatever unit turns its measurement into “score points,” so every term is in the same unit before anything is added. Fitting good weights is what regression does; here we only use them.

Notebook U2-2_Vectors-2_DotProductAndAngle, §1, the dot product by hand and with `np.dot`

4.2 Careful: Which Multiplication?

“Multiply two vectors” is ambiguous, because there are several different products. The **symbol** decides which one is meant:

Written	Name	Gives
$c \bar{v}$	scalar multiplication	a vector (each entry times c)
$\bar{v} \cdot \bar{w}$	dot product	a number
$\bar{v} \times \bar{w}$	cross product	a vector , only in $\mathbb{R}^3$ (not in this course)
$\bar{v} \bar{w}$	side by side	undefined for two column vectors

Be careful with the last row. Writing two vectors side by side, $\bar{v} \bar{w}$, is **not** the dot product; always write the dot. Side by side does get a meaning in Unit 2-3, for matrices.

The same trap exists in code. With the emails stored as numpy arrays `d1` and `d3`:

- `d1 @ d3` is the dot product: 6.
- `d1 * d3` is **not**: it multiplies entry by entry and stops there, giving `[4 0 2 0]`. Adding those four entries up would give the dot product; `*` skips the adding.

In other words, the dot product is “multiply entry by entry, *then add*”: `(d1 * d3).sum()` and `np.dot(d1, d3)` both give 6, the same as `d1 @ d3`. With plain Python lists neither works: `[2, 1, 2, 0] * [2, 0, 1, 2]` raises a `TypeError`, because Python does not know how to multiply two lists.

Exercise 4.1

Let $\bar{u}, \bar{v}, \bar{w}$ be vectors in $\mathbb{R}^2$. For each expression, say whether it is a **number**, a **vector**, or **undefined**.

1. $3\bar{v}$
2. $\bar{v} \cdot \bar{w}$
3. $(\bar{v} \cdot \bar{w}) \bar{u}$
4. $(\bar{v} \cdot \bar{w}) \cdot \bar{u}$
5. $\|\bar{v}\| \bar{w}$
6. $\bar{v} + (\bar{v} \cdot \bar{w})$
7. $\bar{v} \bar{w}$

Solution to Exercise 4.1

Work from the inside out, replacing each piece by what it produces.

1. A number times a vector: a **vector**.
2. A dot product: a **number**.
3. The bracket is a number, and a number times $\bar{u}$ is a **vector**: $\bar{u}$ stretched by the amount $\bar{v} \cdot \bar{w}$. (Projection, later in this section, has exactly this shape.)
4. **Undefined**. The bracket is a number, and a dot product needs a vector on each side.
5. $\|\bar{v}\|$ is a number, so this is a **vector**.
6. **Undefined**. A vector plus a number is not defined (numpy would quietly broadcast it, but that is not vector algebra).
7. **Undefined** for two column vectors, as the table says.

4.3 Rules, and What the Sign Tells You

The dot product obeys rules much like ordinary multiplication.

Properties of the dot product

- $\bar{v} \cdot \bar{w} = \bar{w} \cdot \bar{v}$
- $\bar{u} \cdot (\bar{v} + \bar{w}) = \bar{u} \cdot \bar{v} + \bar{u} \cdot \bar{w}$
- $(c\bar{v}) \cdot \bar{w} = c(\bar{v} \cdot \bar{w})$
- $\bar{v} \cdot \bar{v} = \|\bar{v}\|^2 \geq 0$

In words: order does not matter, the dot product distributes over a sum, and a scalar can be pulled out front. Each rule is inherited from ordinary numbers, one component at a time. For instance, with $\bar{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\bar{v} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$, and $\bar{w} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$:

$$\begin{aligned}\bar{u} \cdot (\bar{v} + \bar{w}) &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 + 8 = 10, & \bar{u} \cdot \bar{v} + \bar{u} \cdot \bar{w} &= (3 + 0) + (-1 + 8) = 3 + 7 = 10, \\ (2\bar{u}) \cdot \bar{v} &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 6, & 2(\bar{u} \cdot \bar{v}) &= 2 \times 3 = 6.\end{aligned}$$

The last rule is the link to **length**. Dotting a vector with itself multiplies each component by itself and adds, $\bar{v} \cdot \bar{v} = v_1^2 + v_2^2 + \cdots + v_n^2$, which is exactly the square of the norm. So

$$\|\bar{v}\| = \sqrt{\bar{v} \cdot \bar{v}}.$$

For the plane, $\bar{v} \cdot \bar{v} = 150^2 + 200^2 = 22500 + 40000 = 62500$, and $\sqrt{62500} = 250$ as before.

Watch out: $\bar{v} \cdot \bar{v}$ is the length squared

$\bar{v} \cdot \bar{v} = 62500$ is not the plane's speed; it is the speed *squared*. Forgetting the final square root is one of the most common slips in this unit. A quick sanity check catches it: a length should

be in the same units as the components (m/s), while $\bar{v} \cdot \bar{v}$ is in (m/s)².

Even before we compute any angles, the **sign** of a dot product is a verdict on direction:

- **Positive:** the vectors broadly **agree**, less than a quarter turn (90°) apart.
- **Zero:** they are **perpendicular**. Neither has any component along the other.
- **Negative:** they broadly **oppose**, more than a quarter turn apart.

The next subsection shows why: the dot product always has the same sign as the cosine of the angle between the vectors.

Running example: a plane flying along $\bar{v}$, in a wind $\bar{w}$.

- $\bar{v} \cdot \bar{w} > 0$: the wind is partly a **tailwind**, pushing the plane along. $\bar{v} \cdot \bar{w} < 0$: it is partly a **headwind**, holding the plane back.
- $\bar{v} \cdot \bar{w} = 0$: a pure **crosswind**. There is no gain or loss along the heading, but the wind still pushes the plane **sideways**.

Worked example: three winds, one plane.

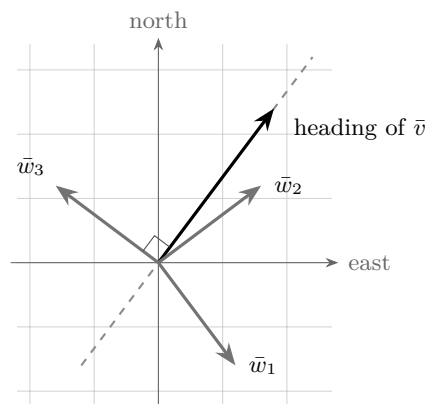
Our plane flies with $\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$. Here are three winds, each blowing at 50 m/s, in different directions:

$$\bar{w}_1 = \begin{bmatrix} 30 \\ -40 \end{bmatrix} : \quad \bar{v} \cdot \bar{w}_1 = 4500 - 8000 = -3500$$

$$\bar{w}_2 = \begin{bmatrix} 40 \\ 30 \end{bmatrix} : \quad \bar{v} \cdot \bar{w}_2 = 6000 + 6000 = 12000$$

$$\bar{w}_3 = \begin{bmatrix} -40 \\ 30 \end{bmatrix} : \quad \bar{v} \cdot \bar{w}_3 = -6000 + 6000 = 0$$

So $\bar{w}_1$ is partly a **headwind**, $\bar{w}_2$ partly a **tailwind**, and $\bar{w}_3$ a pure **crosswind**.



Three winds of 50 m/s against one heading.

Look back at Section 3.2: the wind there was $\bar{w}_1$, and the plane's ground speed fell from 250 to about 241 m/s even though $\bar{w}_1$ pushes partly east. The negative dot product predicted that: taken as a whole, $\bar{w}_1$ opposes the heading. One number, computed with two multiplications and an addition, settles a question that was not obvious from the picture.

4.4 The Angle Between Two Vectors

The size of a dot product grows when either vector gets longer. If we divide out both lengths, what is left depends only on the directions, and it turns out to be the cosine of the angle between them.

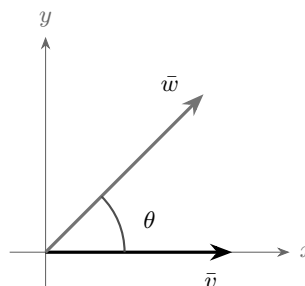
The angle between two vectors

For nonzero $\vec{v}$ and $\vec{w}$, the angle θ between them satisfies

$$\cos \theta = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}$$

- Dividing by both lengths leaves **direction alone**.
- The result is always between -1 and 1 , as a cosine must be.

A reminder about cosine, since it does all the work here. In a right triangle, the cosine of an angle is the side *next to* the angle divided by the hypotenuse. Three values are worth knowing by heart: $\cos 0^\circ = 1$ (same direction), $\cos 90^\circ = 0$ (perpendicular), and $\cos 180^\circ = -1$ (opposite directions). To turn a cosine back into an angle, use the inverse cosine, written $\cos^{-1}$ or **arccos**, on a calculator.



Why the formula works. Take the easiest case, where $\vec{w}$ lies along the x -axis, say $\vec{w} = \begin{bmatrix} a \\ 0 \end{bmatrix}$ with $a > 0$. Then $\vec{v} \cdot \vec{w} = a v_1$ and $\|\vec{w}\| = a$, so the formula's right-hand side is $a v_1 / (\|\vec{v}\| a) = v_1 / \|\vec{v}\|$. Now drop a perpendicular from the head of $\vec{v}$ to the x -axis: the right triangle has hypotenuse $\|\vec{v}\|$ and side v_1 next to the angle θ , so $\cos \theta = v_1 / \|\vec{v}\|$ as well. The two agree. Any other pair of vectors can be turned together until $\vec{w}$ lies along the x -axis; turning changes neither the lengths nor the angle, and (it can be shown) not the dot product either, so the formula holds for every pair. The same picture explains the sign rule: if θ is more than 90° , the head of $\vec{v}$ lies to the *left* of the y -axis, v_1 is negative, and so is the dot product.

Procedure: the angle between two vectors

1. Compute the dot product $\vec{v} \cdot \vec{w}$.
2. Compute both lengths, $\|\vec{v}\|$ and $\|\vec{w}\|$.
3. Divide: $\cos \theta = \vec{v} \cdot \vec{w} / (\|\vec{v}\| \|\vec{w}\|)$, a number between -1 and 1 .
4. Undo the cosine: $\theta = \cos^{-1}(\cos \theta)$, an angle between 0° and 180° .

Worked example: a plane heading east, wind blowing northeast. Take $\vec{v}$ pointing east and $\vec{w}$ pointing northeast:

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} : \quad \vec{v} \cdot \vec{w} = (1)(1) + (0)(1) = 1, \quad \|\vec{v}\| = 1, \quad \|\vec{w}\| = \sqrt{2}.$$

So

$$\cos \theta = \frac{1}{1 \cdot \sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707 \Rightarrow \theta = 45^\circ.$$

Northeast is halfway between east and north, so 45° is exactly what the picture predicts.

The formula keeps working where no picture can follow:

- **Email:** for $\vec{d}_1$ and $\vec{d}_3$, $\cos \theta = 6/(3 \cdot 3) = 2/3$, so $\theta \approx 48^\circ$. The two emails share some words but not the same mix, so they are neither parallel nor perpendicular. The long spam gives the same angle with $\vec{d}_3$: $\vec{d}_2 \cdot \vec{d}_3 = 8 + 0 + 4 + 0 = 12$ and $12/(6 \cdot 3) = 2/3$ again, because $\vec{d}_2$ points the same way as $\vec{d}_1$.
- **Patient:** there is no picture in $\mathbb{R}^{40}$, but the formula still returns a number in $[-1, 1]$, so “the angle between two patients” makes sense even though no one can draw it.

Worked example: two vectors that nearly agree. For $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$: the dot product is 11, the lengths are $\sqrt{5}$ and 5, so

$$\cos \theta = \frac{11}{5\sqrt{5}} \approx 0.984, \quad \theta \approx 10.3^\circ.$$

A cosine close to 1 means a small angle: these two arrows point almost the same way.

Worked example: the plane and its three winds. With $\|\vec{v}\| = 250$ and every wind of length 50, the denominator is $250 \times 50 = 12500$ each time:

Wind	$\vec{v} \cdot \vec{w}$	$\cos \theta$	θ	Reading
$\vec{w}_1 = \begin{bmatrix} 30 \\ -40 \end{bmatrix}$	-3500	-0.28	$\approx 106.3^\circ$	a little past perpendicular: partly headwind
$\vec{w}_2 = \begin{bmatrix} 40 \\ 30 \end{bmatrix}$	12000	0.96	$\approx 16.3^\circ$	nearly along the heading: mostly tailwind
$\vec{w}_3 = \begin{bmatrix} -40 \\ 30 \end{bmatrix}$	0	0	90°	exactly across: pure crosswind

Worked example: an angle in $\mathbb{R}^3$. For $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$: the dot product is $0 + 0 + 1 = 1$, each length is $\sqrt{2}$, so $\cos \theta = 1/(\sqrt{2}\sqrt{2}) = 1/2$ and $\theta = 60^\circ$.

Watch out: degrees versus radians

`np.arccos` (and most programming languages) returns the angle in **radians**, a different unit in which a full turn is $2\pi \approx 6.28$ instead of 360° . `np.arccos(2/3)` returns about 0.841, which is the email angle in radians; `np.degrees(np.arccos(2/3))` converts it to about 48.19° . On a calculator, check whether it is in degree or radian mode before trusting an answer.

Watch out: a cosine a hair above 1

Mathematically $\cos \theta$ is never above 1. In floating point (Unit 1-5) it can be, by one rounding step. For $\vec{v} = \begin{bmatrix} 0.1 \\ 0.7 \end{bmatrix}$ compared with itself, `(v @ v) / (np.linalg.norm(v) * np.linalg.norm(v))` evaluates to 1.0000000000000002. Then `np.arccos` of it returns `nan` (with a warning, “invalid value encountered in arccos”), and Python’s `math.acos` stops with a “math domain error.” The standard fix is to clip the cosine into range first: `np.arccos(np.clip(c, -1, 1))`. And if a hand computation gives a cosine of, say, 1.3, that is not rounding: it is an arithmetic mistake, so recheck the dot product and the lengths.

4.5 Cosine Similarity

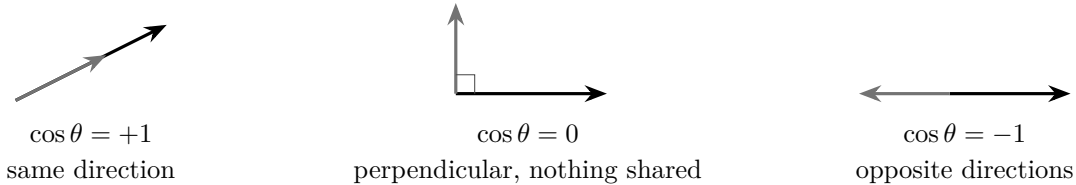
The angle formula is used so widely in data science that it has a name of its own.

Definition: Cosine similarity

The **cosine similarity** of nonzero $\vec{v}, \vec{w}$ is the cosine of the angle between them:

$$\cos \theta = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}$$

It runs from +1 to -1, and three values are worth picturing:



Why does it deserve its own name?

- It sees **direction only**: scaling either vector by a **positive** number leaves it unchanged. Doubling $\vec{v}$, for instance, doubles $\vec{v} \cdot \vec{w}$ and also doubles $\|\vec{v}\|$, and the two doublings cancel in the fraction. (Scaling by a negative number reverses the vector, and so flips the sign of the cosine.)
- That is exactly right when size is noise, like the **length** of an email. A long email and a short one about the same thing should count as alike.

A second way to compute it follows from the rule that scalars pull out of a dot product: dividing by the two lengths is the same as normalizing both vectors first, so

$$\cos \theta = \frac{\vec{v}}{\|\vec{v}\|} \cdot \frac{\vec{w}}{\|\vec{w}\|} = \hat{v} \cdot \hat{w}.$$

Cosine similarity is the dot product of the two unit vectors. For the emails,

$$\hat{d}_1 \cdot \hat{d}_3 = \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 2/3 \\ 0 \\ 1/3 \\ 2/3 \end{bmatrix} = \frac{4}{9} + 0 + \frac{2}{9} + 0 = \frac{2}{3},$$

the same answer as before. Search systems use this: they normalize every document once, store the unit vectors, and then each comparison is a single dot product.

Worked example: patients who use care the same way. Record each patient's visits in a year as (emergency, primary care, specialist):

$$\bar{a}_1 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}, \quad \bar{a}_2 = \begin{bmatrix} 2 \\ 8 \\ 4 \end{bmatrix}, \quad \bar{a}_3 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}.$$

Which patient is most like $\bar{a}_1$? The lengths are $\|\bar{a}_1\| = \sqrt{21}$, $\|\bar{a}_2\| = \sqrt{84} = 2\sqrt{21}$, and $\|\bar{a}_3\| = \sqrt{11}$.

- **Cosine:** $\bar{a}_2 = 2\bar{a}_1$, so their cosine is exactly 1. For $\bar{a}_3$: $\bar{a}_1 \cdot \bar{a}_3 = 3 + 4 + 2 = 9$ and $\cos \theta = 9/(\sqrt{21}\sqrt{11}) = 9/\sqrt{231} \approx 0.59$. Cosine picks $\bar{a}_2$.

- **Distance:** subtract and take lengths,

$$\bar{a}_2 - \bar{a}_1 = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}, \text{ length } \sqrt{21} \approx 4.58; \quad \bar{a}_3 - \bar{a}_1 = \begin{bmatrix} 2 \\ -3 \\ -1 \end{bmatrix}, \text{ length } \sqrt{14} \approx 3.74.$$

Distance picks $\bar{a}_3$.

The two measures answer different questions. Patient $\bar{a}_2$ uses care in exactly the same *pattern* as $\bar{a}_1$, mostly primary care, just twice as much; patient $\bar{a}_3$ makes about as many visits in total but leans on the emergency department. If the question is “who uses the health system the same way?”, cosine is right. If it is “who has a similar total burden?”, distance is closer to it.

Worked example: a negative cosine. Counts are never negative, so the dot product of two count vectors is never negative, and cosine similarity between two emails always lies between 0 and 1: emails can be unrelated but never “opposite.” Standardized measurements are different, since they are negative whenever a value is below average. Take two standardized patient profiles, each above average exactly where the other is below:

$$\bar{z} = \begin{bmatrix} 1.0 \\ 0.5 \\ -0.5 \end{bmatrix}, \quad -\bar{z} = \begin{bmatrix} -1.0 \\ -0.5 \\ 0.5 \end{bmatrix}, \quad \cos \theta = \frac{\bar{z} \cdot (-\bar{z})}{\|\bar{z}\| \|\bar{z}\|} = \frac{-\|\bar{z}\|^2}{\|\bar{z}\|^2} = -1,$$

the most dissimilar two profiles can be.

Watch out: similarity goes up, distance goes down

A *larger* cosine similarity means *more* alike, while a *smaller* distance means more alike. Mixing the two up reverses every ranking. To make cosine behave like a distance, some software reports the **cosine distance** $1 - \cos \theta$ instead, which is 0 for identical directions; read the documentation to see which one a function returns.

Notebook U2-2_Vectors-3_ProjectionAndSimilarity, §3, a patient similarity heatmap

4.6 Why Spam Filters Compare by Cosine

Here are the three emails from Section 2.3 again, with their count vectors and lengths:

$\bar{d}_1$ “Win cash now! Free cash, free entry.” **spam**
 $\bar{d}_2$ “Win cash now! Free cash, free entry. Win free cash, free prizes, cash today.” **spam**
 $\bar{d}_3$ “Free lunch at the meeting. After the meeting, free parking; bring cash.” **not spam**

	$\bar{d}_1$	$\bar{d}_2$	$\bar{d}_3$
<i>free</i>	2	4	2
<i>win</i>	1	2	0
<i>cash</i>	2	4	1
<i>meeting</i>	0	0	2

$$\|\bar{d}_1\| = \|\bar{d}_3\| = 3, \quad \|\bar{d}_2\| = 6$$

Which email is **most like** the spam $\bar{d}_1$? The two measures we have disagree:

- **Distance** picks $\bar{d}_3$, which is not spam: it is fooled by **length**.
- **Cosine** picks $\bar{d}_2$: the same spam, only longer.

	Distance from $\bar{d}_1$	Cosine with $\bar{d}_1$
$\bar{d}_2$ (spam, longer)	$\sqrt{2^2 + 1^2 + 2^2 + 0^2} = 3$	$(8 + 2 + 8 + 0)/(3 \cdot 6) = 1$
$\bar{d}_3$ (not spam)	$\sqrt{0^2 + 1^2 + 1^2 + 2^2} \approx 2.45$	$(4 + 0 + 2 + 0)/(3 \cdot 3) \approx 0.67$

Here is why distance goes wrong. The difference $\bar{d}_2 - \bar{d}_1$ is $\bar{d}_1$ itself, so $\bar{d}_2$ sits a full $\|\bar{d}_1\| = 3$ away, purely because it is twice as long. The email $\bar{d}_3$ happens to be the same length as $\bar{d}_1$, so it lands closer ($\sqrt{6} \approx 2.45$) even though it is a different kind of email. Cosine ignores length altogether: $\bar{d}_2 = 2\bar{d}_1$ points in exactly the same direction, so it scores 1, the highest possible, while $\bar{d}_3$ scores only $2/3$. For comparing documents, where length is noise, cosine gives the right answer.

Worked example: a short spam arrives. Suppose a filter has only two labeled emails to compare against: the long spam $\bar{d}_2$ and the meeting email $\bar{d}_3$. A new email arrives: “Win free cash!”, with count vector and length

$$\bar{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \quad \|\bar{x}\| = \sqrt{3}.$$

The filter labels it like whichever known email it is most similar to.

- **By distance:**

$$\bar{x} - \bar{d}_2 = \begin{bmatrix} -3 \\ -1 \\ -3 \\ 0 \end{bmatrix}, \text{ length } \sqrt{19} \approx 4.36; \quad \bar{x} - \bar{d}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \\ -2 \end{bmatrix}, \text{ length } \sqrt{6} \approx 2.45.$$

Nearest is $\bar{d}_3$: **not spam**. Wrong.

- **By cosine:** $\bar{x} \cdot \bar{d}_2 = 4 + 2 + 4 + 0 = 10$, so $\cos \theta = 10/(6\sqrt{3}) \approx 0.96$; $\bar{x} \cdot \bar{d}_3 = 2 + 0 + 1 + 0 = 3$, so $\cos \theta = 3/(3\sqrt{3}) \approx 0.58$. Most similar is $\bar{d}_2$: **spam**. Right.

A short email is close to other short emails and far from long ones, whatever it says. Distance measures “how much text” as much as “which words”; cosine measures only which words, in what proportions.

Worked example: normalize first, and distance agrees. If length is the problem, remove it: normalize every email, then measure distance between the unit vectors. Since $\hat{d}_2 = \hat{d}_1$, their distance is 0. And

$$\hat{d}_1 - \hat{d}_3 = \begin{bmatrix} 2/3 - 2/3 \\ 1/3 - 0 \\ 2/3 - 1/3 \\ 0 - 2/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1/3 \\ 1/3 \\ -2/3 \end{bmatrix}, \quad \|\hat{d}_1 - \hat{d}_3\| = \sqrt{0 + \frac{1}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{\frac{6}{9}} \approx 0.82.$$

Now distance also calls $\bar{d}_2$ the closest. This is no accident. For any two unit vectors, the rules of the dot product give

$$\|\hat{v} - \hat{w}\|^2 = (\hat{v} - \hat{w}) \cdot (\hat{v} - \hat{w}) = \hat{v} \cdot \hat{v} - 2\hat{v} \cdot \hat{w} + \hat{w} \cdot \hat{w} = 1 - 2\cos\theta + 1 = 2 - 2\cos\theta,$$

so once lengths are removed, a bigger cosine always means a smaller distance, and the two measures rank everything the same way. (Check: $2 - 2 \cdot \frac{2}{3} = \frac{2}{3}$, and $\sqrt{2/3} \approx 0.82$.)

Exercise 4.2

Take the email $\bar{e}$ from Exercise 2.2 (“Meeting at noon: free lunch, and bring cash for the free coffee”), with counts 2, 0, 1, 1 of *free*, *win*, *cash*, *meeting*. Compute its cosine similarity with $\bar{d}_1$ and with $\bar{d}_3$. If a filter labels a new email like the known email it is most similar to, what label does $\bar{e}$ get?

Solution to Exercise 4.2

First the length: $\|\bar{e}\| = \sqrt{4 + 0 + 1 + 1} = \sqrt{6}$. Both $\bar{d}_1$ and $\bar{d}_3$ have length 3.

- With $\bar{d}_1$: $\bar{e} \cdot \bar{d}_1 = 4 + 0 + 2 + 0 = 6$, so $\cos\theta = 6/(3\sqrt{6}) \approx 0.82$.
- With $\bar{d}_3$: $\bar{e} \cdot \bar{d}_3 = 4 + 0 + 1 + 2 = 7$, so $\cos\theta = 7/(3\sqrt{6}) \approx 0.95$.

The email is most similar to $\bar{d}_3$, so it is labeled **not spam**, matching the impression by eye in Exercise 2.2. The deciding word is *meeting*: it is the only word that $\bar{e}$ shares with $\bar{d}_3$ and not with $\bar{d}_1$, and it adds $1 \times 2 = 2$ to the dot product with $\bar{d}_3$.

4.7 Beyond Counting Words: Embeddings

Count vectors have a blind spot:

- “Win free cash now” and “Claim complimentary money today” mean the **same thing**.
- But they share **no words**, so their count vectors are perpendicular: cosine 0.
- Counts see **spelling**, not **meaning**.

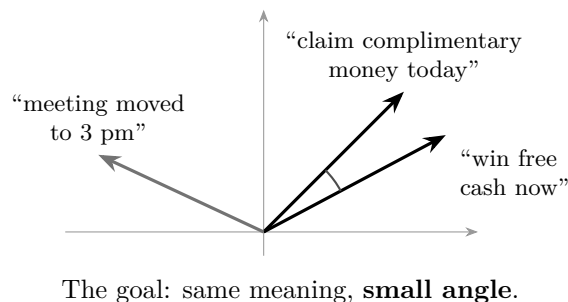
Worked example: the blind spot in numbers. Use the eight-word vocabulary *win*, *free*, *cash*, *now*, *claim*, *complimentary*, *money*, *today*. Call the count vector of “Win free cash now” $\bar{s}_1$ and that of “Claim complimentary money today” $\bar{s}_2$. Stored the way a data table stores records, one sentence per row, they are:

	<i>win</i>	<i>free</i>	<i>cash</i>	<i>now</i>	<i>claim</i>	<i>complimentary</i>	<i>money</i>	<i>today</i>
$\bar{s}_1$	1	1	1	1	0	0	0	0
$\bar{s}_2$	0	0	0	0	1	1	1	1

Wherever one has a 1 the other has a 0, so every product in $\bar{s}_1 \cdot \bar{s}_2$ is 0 and the cosine similarity is 0: as unrelated as two sentences can be, by this measure. The same happens with single words. Over any vocabulary, the count vector of the one-word text “cat” has a single 1 in the *cat* slot, and “dog” has a single 1 in the *dog* slot, so $\text{cat} \cdot \text{dog} = 0$. So is *cat* with *agglomerative*. Counts cannot tell that a cat is more like a dog than like a clustering method.

What we want instead:

- “cat” and “dog” should come out **similar**; “cat” and “agglomerative” should not.
- Counts cannot do it: each word is its own unit vector, so **any** two different words are perpendicular.
- We need vectors built from **meaning**. Those are **embeddings**.



The fix keeps the geometry and changes the vectors. Everything we have built (dot product, length, cosine) stays exactly the same; only the recipe for turning text into numbers changes. The picture on the right is a cartoon of the goal: the two sentences that mean the same thing should point nearly the same way, and an unrelated sentence should point somewhere else.

4.8 What an Embedding Looks Like

The idea: an embedding

- Each word becomes a vector of **learned** numbers, not counts: hundreds to thousands of components.
- A model learns them by **predicting each word from the words around it**.
- Trained so that **similar meanings point the same way**.

Why would predicting words from their neighbors produce meaning? Think of the blank in “The ____ sat on the mat.” Words that fit the same blanks, such as *cat*, *dog*, and *kitten*, are used in similar ways, and a model that must fill blanks well ends up giving them similar vectors. Words that never fit the same blanks, such as *cat* and *agglomerative*, end up pointing in unrelated directions. Nobody tells the model what any word means; the meaning is squeezed out of how words are used across a very large amount of text.

Here is what that looks like for one sentence, “The cat sat on the mat,” with one column per word:

	<i>the</i>	<i>cat</i>	<i>sat</i>	<i>on</i>	<i>the</i>	<i>mat</i>
entry 1	0.23	0.32	0.23	−2.63	0.23	0.01
2	0.51	−1.47	0.00	−0.77	0.51	1.99
⋮	⋮	⋮	⋮	⋮	⋮	⋮
$n - 1$	−0.21	0.86	−1.47	0.23	−0.21	0.23
n	1.99	0.10	0.51	0.11	1.99	0.50

How to read the table:

- Each column is one word’s **vector**. It has n entries, where n is the size of the embedding (typically a few hundred); the dots stand for the entries not shown.
- The individual entries have no meaning on their own. Entry 2 does not measure “animal-ness”

or anything else you could name; only the vector as a whole, compared with other vectors by cosine, carries meaning.

- Both copies of *the* get the **same** column: the vector belongs to the word, not to its position in the sentence. (That is true of the embeddings described here. Newer language models go one step further and adjust a word's vector to fit the sentence around it, but the starting point is a table like this one.)
- Side by side, the columns form a grid of numbers, a **matrix**: Unit 2-3.

Worked example: toy embeddings. Real embeddings are too long to work by hand, so here are made-up three-component ones, invented only to show the arithmetic:

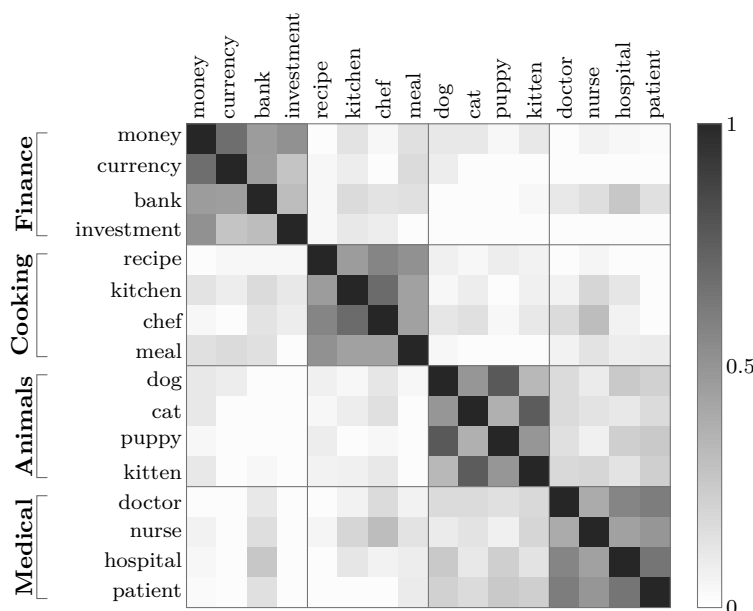
$$\text{cat} = \begin{bmatrix} 0.8 \\ 0.6 \\ 0.1 \end{bmatrix}, \quad \text{dog} = \begin{bmatrix} 0.7 \\ 0.7 \\ 0.2 \end{bmatrix}, \quad \text{agglomerative} = \begin{bmatrix} 0.1 \\ -0.2 \\ 0.9 \end{bmatrix}.$$

- *cat* and *dog*: the dot product is $0.56 + 0.42 + 0.02 = 1.00$; the lengths are $\sqrt{1.01} \approx 1.005$ and $\sqrt{1.02} \approx 1.010$; so $\cos \theta \approx 1.00 / (1.005 \times 1.010) \approx 0.985$, an angle of about 10° .
- *cat* and *agglomerative*: the dot product is $0.08 - 0.12 + 0.09 = 0.05$; the second length is $\sqrt{0.86} \approx 0.927$; so $\cos \theta \approx 0.05 / (1.005 \times 0.927) \approx 0.054$, an angle of about 87° , nearly perpendicular.

This is exactly the behavior counts could not produce: two different words, yet a cosine near 1.

4.9 Cosine Similarity Between Word Embeddings

The heatmap below does that computation for 16 real words, four from each of four topics. Each cell is the cosine similarity of two words' embeddings, shaded from white (0) to dark (1).



Cosine similarity between the embeddings of 16 words. Rows and columns list the same words in the same order.

How to read it:

- Each cell is the **cosine similarity** of two words' embeddings; **darker** is more similar. To look up *cat* and *kitten*, go along the *cat* row to the *kitten* column: the cell is dark, a cosine of 0.75.
- The diagonal is 1: every word with itself. A vector always makes an angle of 0° with itself.
- **Dark 4×4 blocks:** words on the same topic point nearly the same way, a **small angle**.
- **Light elsewhere:** words on different topics are **nearly perpendicular**. *cat* and *money* have cosine 0.11, an angle of about 84° .

The model was never told the topics. The blocks come only from how the words are used in text.

Read honestly, the blocks are a strong tendency rather than a law. The lightest cell inside a block, *currency* with *investment* at 0.27, is a shade lighter than the darkest cell outside the blocks, *chef* with *nurse* at 0.30. Similarity from usage is powerful, but it is about how words are used, not a dictionary's categories.

Exercise 4.3

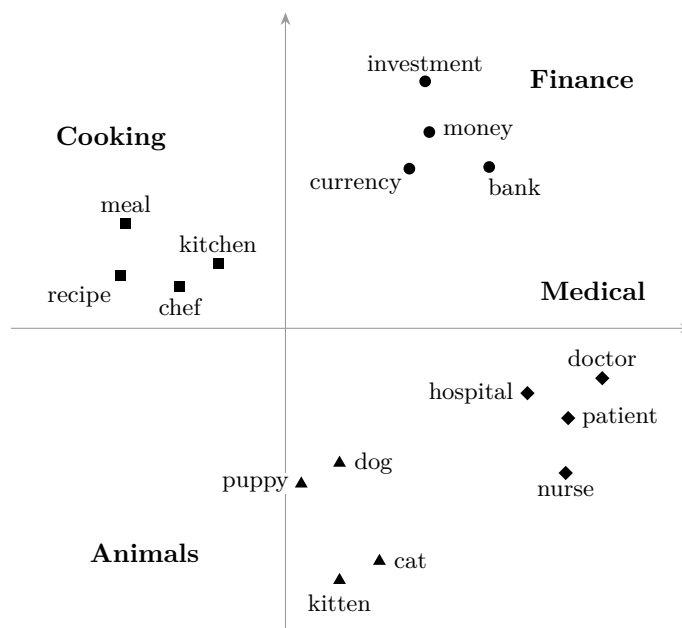
Use the heatmap.

1. Apart from *doctor* itself, which word is most similar to *doctor*?
2. The heatmap is symmetric: the cell in row *dog*, column *puppy* equals the cell in row *puppy*, column *dog*. Why must that be so?
3. Convert the cosines for *cat* with *kitten* (0.75) and *doctor* with *patient* (0.60) into angles.

Solution to Exercise 4.3

1. Read along the *doctor* row: within the Medical block the values are 0.39 (*nurse*), 0.57 (*hospital*), and 0.60 (*patient*), and everything outside the block is lighter. The most similar word is ***patient***.
2. Because the dot product does not care about order, $\vec{v} \cdot \vec{w} = \vec{w} \cdot \vec{v}$, and the lengths in the denominator are the same two numbers either way. So the cosine of *dog* with *puppy* is the cosine of *puppy* with *dog*.
3. $\cos^{-1}(0.75) \approx 41.4^\circ$ and $\cos^{-1}(0.60) \approx 53.1^\circ$. Even the most similar pairs here are tens of degrees apart; what matters is that they are much closer than the 80° to 90° of words on unrelated topics.

4.10 Similar Meanings Cluster Together



The same 16 embeddings, flattened to 2-D. Each topic has its own marker shape.

What the picture shows:

- Each word's hundreds of numbers are **flattened to 2** so we can draw them, using **PCA** (the end of this unit) plus a method called t-SNE. Flattening necessarily throws information away; these methods are designed to keep, as far as possible, which points are near which.
- Each topic lands in its own **cluster**: similar meaning, nearby points. The heatmap's four dark blocks and the picture's four clusters are the same fact, seen two ways.
- The axes carry no units; only **who is near whom** matters. It would be meaningless to say that *bank* has an *x*-coordinate of about 25.

The heatmap speaks of *similarity* (cosine) and this picture of *nearness* (distance). The worked example at the end of Section 4.6 showed why they tell the same story: once vectors are normalized, $\|\hat{v} - \hat{w}\|^2 = 2 - 2\cos\theta$, so a large cosine and a small distance are the same thing.

Back to the spam filter. Whole sentences get embeddings too. Now “win free cash now” and “claim complimentary money today” point nearly the **same way**, and the **same cosine** finds it. Search engines rank results this way: they embed your query, embed every document, and return the documents whose vectors make the smallest angle with the query's. The same idea finds clinical notes that describe the same symptoms in different words, which is exactly what the count vectors of Section 2.3 could not do.

4.11 Orthogonality

A dot product of exactly zero is important enough to get its own name.

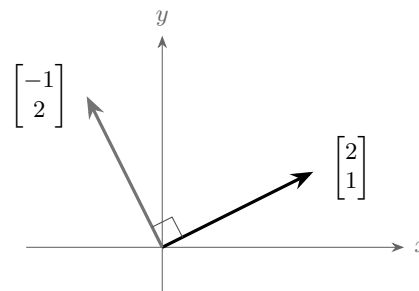
Definition: Orthogonal and orthonormal

- $\bar{v}$ and $\bar{w}$ are **orthogonal** (perpendicular) if $\bar{v} \cdot \bar{w} = 0$.
- They are **orthonormal** if they are orthogonal **and** both are unit vectors, $\|\bar{v}\| = \|\bar{w}\| = 1$.

- **Why** zero means perpendicular: $\cos 90^\circ = 0$, so by the angle formula a dot product of 0 between nonzero vectors means a 90° angle.

- **Check** the pair in the picture:

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = (2)(-1) + (1)(2) = -2 + 2 = 0.$$



- They are **not** orthonormal, since each has length $\sqrt{2^2 + 1^2} = \sqrt{5}$. Normalize both and they are, just like $\hat{i}$ and $\hat{j}$ (worked out below).
- The test needs no square roots and no trigonometry, only multiplying and adding, so it is **cheap** even in $\mathbb{R}^{500}$.

Normalizing the pair in the picture divides each by $\sqrt{5}$:

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.894 \\ 0.447 \end{bmatrix}, \quad \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \approx \begin{bmatrix} -0.447 \\ 0.894 \end{bmatrix}.$$

Each now has length 1, and their dot product is still 0 (scaling cannot turn a zero dot product into anything else), so the pair is orthonormal: a copy of $\hat{i}$ and $\hat{j}$, turned.

In the running examples:

- **Velocity:** a pure **crosswind** is orthogonal to the plane's heading. It adds nothing along the heading; it only pushes the plane sideways. The wind $\bar{w}_3 = \begin{bmatrix} -40 \\ 30 \end{bmatrix}$ of Section 4.3 is one.
- **Email:** two emails with **no words in common** have dot product 0, so they are orthogonal. "Meeting at noon" has counts 0, 0, 0, 1, and its dot product with $\bar{d}_1$ (counts 2, 1, 2, 0) is $0 + 0 + 0 + 0 = 0$.

Worked example: a pair in $\mathbb{R}^3$. Are $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$ orthogonal? The dot product is $2 - 4 + 2 = 0$: yes. Each has length $\sqrt{1 + 4 + 4} = 3$, so they are not orthonormal, but dividing each by 3 makes them so:

$$\begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}.$$

Worked example: building a perpendicular vector. In the plane there is a quick trick: **swap the two components and change one sign**. The vector $\begin{bmatrix} -b \\ a \end{bmatrix}$ is always perpendicular to $\begin{bmatrix} a \\ b \end{bmatrix}$, because $a(-b) + b(a) = 0$. So $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$ is perpendicular to $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$. For the plane, the heading $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ gives the crosswind direction $\begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix}$, and a 50 m/s crosswind is $50 \begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix} = \begin{bmatrix} -40 \\ 30 \end{bmatrix}$, which is $\bar{w}_3$.

Worked example: solving for orthogonality. For which k is $\begin{bmatrix} 2 \\ k \end{bmatrix}$ orthogonal to $\begin{bmatrix} 3 \\ -6 \end{bmatrix}$? Set the dot product to zero: $6 - 6k = 0$, so $k = 1$. Check: $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -6 \end{bmatrix} = 6 - 6 = 0$.

Watch out: perpendicular is not opposite

It is tempting to think of orthogonal vectors as “opposite.” They are not: opposite vectors point in reverse directions and have cosine -1 (a dot product as negative as it can be), while orthogonal vectors are a quarter turn apart and have cosine 0 . For data, orthogonal means “nothing in common”: $\bar{d}_1$ and “Meeting at noon” share no words, which is different from being anti-spam. One more corner case: by the definition, the zero vector is orthogonal to *every* vector, since $\bar{0} \cdot \bar{w} = 0$, even though it has no direction and no angle.

4.12 Exercise: Dot Products and Angles

Exercise 4.4

Let

$$\bar{a} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}, \quad \bar{b} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \bar{c} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}.$$

1. Compute $\bar{a} \cdot \bar{b}$ and the angle between $\bar{a}$ and $\bar{b}$.
2. Compute $\bar{a} \cdot \bar{c}$. What does the answer say about them?
3. Compute $\bar{b} \cdot \bar{c}$. What does the **sign** say?
4. Turn $\bar{b}$ into a unit vector.

Solution to Exercise 4.4

1. Multiply matching components and add: $\bar{a} \cdot \bar{b} = (3)(3) + (0)(3) = 9$. For the angle we also need both lengths, $\|\bar{a}\| = \sqrt{9+0} = 3$ and $\|\bar{b}\| = \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$. Then

$$\cos \theta = \frac{9}{3 \cdot 3\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \text{so} \quad \theta = 45^\circ.$$

This is the answer the picture predicts: $\bar{a}$ lies along the x -axis and $\bar{b}$ on the diagonal.

2. $\bar{a} \cdot \bar{c} = (3)(0) + (0)(-2) = 0$, so they are **orthogonal**. One runs along the x -axis and the other along the y -axis.
3. $\bar{b} \cdot \bar{c} = (3)(0) + (3)(-2) = -6$. The sign is **negative**, so the two point more than 90° apart: $\bar{b}$ heads up and to the right while $\bar{c}$ heads straight down. (The angle formula confirms it: $\cos \theta = -6/(3\sqrt{2} \cdot 2) = -1/\sqrt{2}$, so $\theta = 135^\circ$.)
4. Divide $\bar{b}$ by its length, $3\sqrt{2}$:

$$\hat{b} = \frac{1}{3\sqrt{2}} \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}.$$

Check the length: $(1/\sqrt{2})^2 + (1/\sqrt{2})^2 = 0.5 + 0.5 = 1$, as required. $\hat{b}$ points the same way as $\bar{b}$, along the diagonal, but has length 1.

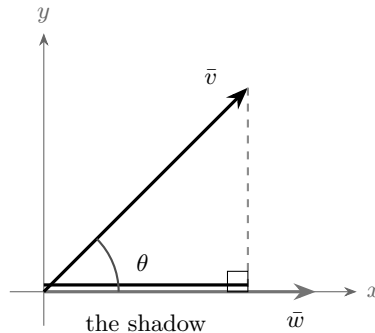
4.13 Projection: the Shadow of One Vector on Another

Imagine a light shining straight down onto the line through $\bar{w}$, at right angles to it. The **shadow** that $\bar{v}$ casts on that line is the part of $\bar{v}$ running **along** $\bar{w}$, and finding it is called projection.

Velocity: a wind $\bar{v}$ blows on a plane heading along $\bar{w}$:

$$\bar{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \hat{w} = \frac{1}{4}\bar{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The wind blows northeast and the plane heads east. How much of the wind helps the plane along, and how much only pushes it sideways?



Projection takes two steps: first a number, then a vector.

Projection in two steps

1. **Dot with $\hat{w}$: a number.** The **component** of $\bar{v}$ along $\bar{w}$, which is the shadow's **length**:

$$\text{comp}_{\bar{w}}(\bar{v}) = \bar{v} \cdot \hat{w} = \|\bar{v}\| \cos \theta$$

2. **Times $\hat{w}$: a vector.** The **projection**, which is that length pointed along $\bar{w}$:

$$\text{proj}_{\bar{w}}(\bar{v}) = (\bar{v} \cdot \hat{w}) \hat{w}$$

Why does step 1 give the shadow's length? In the picture, $\bar{v}$, the shadow, and the dashed drop form a right triangle with $\bar{v}$ as the hypotenuse, so the side along $\bar{w}$ has length $\|\bar{v}\| \cos \theta$. The angle formula of Section 4.4, applied to $\bar{v}$ and $\hat{w}$, says the same thing: $\bar{v} \cdot \hat{w} = \|\bar{v}\| \|\hat{w}\| \cos \theta = \|\bar{v}\| \cos \theta$, because $\|\hat{w}\| = 1$. That is why we dot with $\hat{w}$ rather than $\bar{w}$: a length of 1 drops out.

For the plane:

1. $\bar{v} \cdot \hat{w} = (3)(1) + (3)(0) = 3$: the **tailwind**, how hard the wind pushes along the heading. (As a check, $\|\bar{v}\| \cos \theta = 3\sqrt{2} \cdot \frac{1}{\sqrt{2}} = 3$.)
2. $\text{proj}_{\bar{w}}(\bar{v}) = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$: the tailwind part of the wind, as a vector.

What is left over is the rest of the wind:

$$\bar{v} - \text{proj}_{\bar{w}}(\bar{v}) = \begin{bmatrix} 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix},$$

a **crosswind** pushing the plane sideways. It is orthogonal to the heading, since $\begin{bmatrix} 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \end{bmatrix} = 0 + 0 = 0$; that is the right angle in the picture. So the wind splits cleanly into a part along the heading plus a part at right angles to it.

A version without square roots. Normalizing $\bar{w}$ needs its length, which usually means a square root. The two square roots cancel if we put the steps together: $(\bar{v} \cdot \hat{w}) \hat{w} = \frac{\bar{v} \cdot \bar{w}}{\|\bar{w}\|} \frac{\bar{w}}{\|\bar{w}\|}$, and $\|\bar{w}\|^2 = \bar{w} \cdot \bar{w}$,

so

$$\text{proj}_{\bar{w}}(\bar{v}) = \frac{\bar{v} \cdot \bar{w}}{\bar{w} \cdot \bar{w}} \bar{w}.$$

This is the same projection, just rearranged; use whichever is easier for the numbers in front of you.

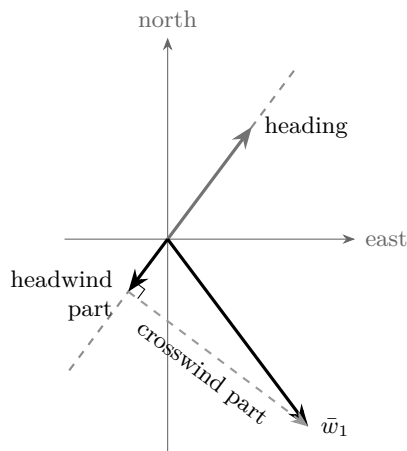
Worked example: projecting onto a diagonal. Project $\bar{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ onto $\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Here $\bar{v} \cdot \bar{w} = 4$ and $\bar{w} \cdot \bar{w} = 2$, so

$$\text{proj}_{\bar{w}}(\bar{v}) = \frac{4}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \quad \text{leftover } \bar{v} - \text{proj}_{\bar{w}}(\bar{v}) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Check that the leftover is orthogonal to $\bar{w}$: $\begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 - 1 = 0$. The component is $\bar{v} \cdot \hat{w} = 4/\sqrt{2} = 2\sqrt{2} \approx 2.83$, which is indeed the length of $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$.

Worked example: the running plane in a headwind. Return to the plane with $\bar{v} = \begin{bmatrix} 150 \\ 200 \end{bmatrix}$, heading $\hat{v} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, and the wind $\bar{w}_1 = \begin{bmatrix} 30 \\ -40 \end{bmatrix}$. Now we project the *wind* onto the *heading*:

1. Component: $\bar{w}_1 \cdot \hat{v} = (30)(0.6) + (-40)(0.8) = 18 - 32 = -14$ m/s. Negative: a 14 m/s **headwind**.
2. Projection: $-14 \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} -8.4 \\ -11.2 \end{bmatrix}$, pointing backward along the heading.
3. Leftover: $\begin{bmatrix} 30 \\ -40 \end{bmatrix} - \begin{bmatrix} -8.4 \\ -11.2 \end{bmatrix} = \begin{bmatrix} 38.4 \\ -28.8 \end{bmatrix}$, the **crosswind**, of size $\sqrt{38.4^2 + 28.8^2} = 48$ m/s.



One unit is 10 m/s.

The leftover is orthogonal to the heading: $(38.4)(0.6) + (-28.8)(0.8) = 23.04 - 23.04 = 0$. And the two parts fit Pythagoras, as the right angle says they must: $14^2 + 48^2 = 196 + 2304 = 2500 = 50^2$, the wind speed squared. So this 50 m/s wind is mostly a crosswind (48 m/s) with a modest headwind (14 m/s). That explains Section 3.2: the ground velocity $\begin{bmatrix} 180 \\ 160 \end{bmatrix}$ has component $180(0.6) + 160(0.8) = 236 = 250 - 14$ m/s along the heading.

Worked example: the spammy part of an email. Projection works in any dimension. Project the meeting email $\bar{d}_3$ onto the direction of the spam $\bar{d}_1$. With $\hat{d}_1 = \frac{1}{3}\bar{d}_1$,

$$\bar{d}_3 \cdot \hat{d}_1 = \frac{6}{3} = 2, \quad \text{proj} = 2\hat{d}_1 = \frac{2}{3} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4/3 \\ 2/3 \\ 4/3 \\ 0 \end{bmatrix}, \quad \text{leftover} = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 4/3 \\ 2/3 \\ 4/3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2/3 \\ -2/3 \\ -1/3 \\ 2 \end{bmatrix}.$$

The projection is the part of $\bar{d}_3$ that looks like spam (its *free* and *cash*), and the leftover is the part that does not, dominated by *meeting*. Check orthogonality: $\frac{4}{3} - \frac{2}{3} - \frac{2}{3} + 0 = 0$. And the lengths fit Pythagoras again: the projection has length 2, the leftover has length $\sqrt{\frac{4}{9} + \frac{4}{9} + \frac{1}{9} + 4} = \sqrt{5}$, and $2^2 + 5 = 9 = \|\bar{d}_3\|^2$.

Watch out: three easy slips in projection

- **Dot with $\hat{w}$, not $\bar{w}$.** Using $\bar{v} \cdot \bar{w}$ as the component makes the answer $\|\bar{w}\|$ times too big. In the first example, $\begin{bmatrix} 3 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \end{bmatrix} = 12$, not the tailwind of 3.
- **Order matters.** Projecting $\bar{v}$ onto $\bar{w}$ gives a vector along $\bar{w}$; projecting $\bar{w}$ onto $\bar{v}$ gives one along $\bar{v}$. They are different vectors.
- **A component can be negative.** It means the shadow points backward along $\bar{w}$, as with the headwind above. Only the *length* of the projection is never negative.

Notebook U2-2_Vectors-3_ProjectionAndSimilarity, §1, the drop drawn with its right angle

4.14 Exercise: Projection and Similarity

Exercise 4.5

1. Project $\bar{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ onto $\bar{w} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$. Sketch it.
2. Two patients are recorded as

$$\bar{p} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \bar{q} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

What is their cosine similarity? Answer without computing anything.

3. Their Euclidean **distance**, on the other hand, is not small. Which of the two measures is the right one, and what does the choice depend on?

Solution to Exercise 4.5

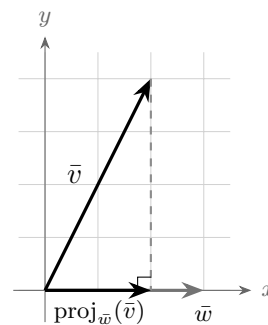
1. Follow the two steps. First normalize $\bar{w}$: $\|\bar{w}\| = 3$, so $\hat{w} = \frac{1}{3} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Then the component and the projection are

$$\bar{v} \cdot \hat{w} = (2)(1) + (4)(0) = 2, \quad \text{proj}_{\bar{w}}(\bar{v}) = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Since $\bar{w}$ lies along the x -axis, the projection just keeps the x -component of $\bar{v}$ and drops the y -component, which is what the sketch shows. The leftover is

$$\bar{v} - \text{proj}_{\bar{w}}(\bar{v}) = \begin{bmatrix} 0 \\ 4 \end{bmatrix},$$

the dashed drop, at right angles to $\bar{w}$.



2. **Exactly** 1. Since $\bar{q} = 3\bar{p}$, the two vectors point along the same line in the same direction, and the angle between them is 0. No arithmetic is needed: scaling a vector by a positive number never changes its direction, and cosine similarity sees direction only.

3. **It depends on whether size carries meaning.** The distance is $\|\bar{q} - \bar{p}\| = \sqrt{2^2 + 4^2} = \sqrt{20} \approx 4.47$, more than twice the length of $\bar{p}$ itself.

- If the components are **counts** (words in a document, visits in a year), then $\bar{q}$ is a bigger version of the same profile, and **cosine** is right.
- If they are **clinical measurements**, where a value three times higher is a genuinely different patient, then **distance** is right.

Choosing between them is a modelling decision about what the numbers mean, not a mathematical one.

4.15 More Practice: Dot Products, Angles, and Projection

Exercise 4.6

Let $\bar{a} = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$ and $\bar{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$.

1. Compute $\bar{a} \cdot \bar{b}$, both lengths, and the angle between them.
2. Project $\bar{a}$ onto $\bar{b}$.

Solution to Exercise 4.6

1. $\bar{a} \cdot \bar{b} = 2 - 2 + 4 = 4$. $\|\bar{a}\| = \sqrt{4 + 1 + 4} = 3$ and $\|\bar{b}\| = \sqrt{1 + 4 + 4} = 3$. So $\cos \theta = 4/9 \approx 0.444$ and $\theta \approx 63.6^\circ$: positive, so less than a quarter turn apart, but not by much.
2. Using the square-root-free form, $\bar{b} \cdot \bar{b} = 9$, so

$$\text{proj}_{\bar{b}}(\bar{a}) = \frac{4}{9} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4/9 \\ 8/9 \\ 8/9 \end{bmatrix} \approx \begin{bmatrix} 0.44 \\ 0.89 \\ 0.89 \end{bmatrix}.$$

Its length is $\frac{4}{9} \times 3 = \frac{4}{3}$, which matches the component $\bar{a} \cdot \hat{b} = 4/3$.

Exercise 4.7

The running plane (heading $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$) meets a wind $\begin{bmatrix} -30 \\ 10 \end{bmatrix}$ m/s. Split the wind into a part along the heading and a crosswind part. Is it a headwind or a tailwind, and how strong is each part?

Solution to Exercise 4.7

Component along the heading: $(-30)(0.6) + (10)(0.8) = -18 + 8 = -10$ m/s, so it is a **headwind** of 10 m/s. The part along the heading is $-10 \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} -6 \\ -8 \end{bmatrix}$, and the crosswind part is the leftover, $\begin{bmatrix} -30 \\ 10 \end{bmatrix} - \begin{bmatrix} -6 \\ -8 \end{bmatrix} = \begin{bmatrix} -24 \\ 18 \end{bmatrix}$, of size $\sqrt{576 + 324} = 30$ m/s. Check: the leftover is orthogonal to the heading, $(-24)(0.6) + (18)(0.8) = -14.4 + 14.4 = 0$, and Pythagoras holds, $10^2 + 30^2 = 1000 = (-30)^2 + 10^2$.

Exercise 4.8

Find both vectors of length 50 that are perpendicular to the heading $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$. What do they mean for the plane?

Solution to Exercise 4.8

Swap and change one sign: $\begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix}$ is perpendicular to $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ and already has length 1, and so does its reverse $\begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix}$. Scaling by 50 gives $\begin{bmatrix} -40 \\ 30 \end{bmatrix}$ and $\begin{bmatrix} 40 \\ -30 \end{bmatrix}$. They are the two pure 50 m/s crosswinds, one pushing the plane to its left (toward the northwest) and one to its right (toward the southeast). Check one: $(-40)(0.6) + (30)(0.8) = -24 + 24 = 0$.

5 Linear Combinations, Span, and Independence

5.1 Linear Combinations

The two operations of Section 3, scaling and adding, combine into the single most important construction in linear algebra.

Definition: Linear combination

A **linear combination** of $\bar{v}_1, \dots, \bar{v}_k$ scales each one and adds:

$$c_1\bar{v}_1 + c_2\bar{v}_2 + \cdots + c_k\bar{v}_k$$

The numbers c_i are the **weights**.

Think of the vectors as ingredients and the weights as a recipe. Two questions matter, and the rest of this section answers them:

1. **Which points** can you reach?
2. For a given point, **which weights** get you there?

We have in fact been doing this all along:

- **Velocity:** $\bar{v} = 150\hat{i} + 200\hat{j}$ is a linear combination with weights 150 and 200.
- **Email:** the word counts are the weights on the word unit vectors (Section 3.7).
- **Patient:** the **average** of $\bar{p}$ and $\bar{q}$ is the linear combination with weights $\frac{1}{2}$ and $\frac{1}{2}$:

$$\frac{1}{2} \begin{bmatrix} 54 \\ 130 \\ 6.1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 58 \\ 138 \\ 6.4 \end{bmatrix} = \begin{bmatrix} 27 + 29 \\ 65 + 69 \\ 3.05 + 3.2 \end{bmatrix} = \begin{bmatrix} 56 \\ 134 \\ 6.25 \end{bmatrix}$$

Worked example: a combination with a negative weight. Compute $2\begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3\begin{bmatrix} 2 \\ -1 \end{bmatrix}$. Scale each vector first, then add:

$$2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} -6 \\ 3 \end{bmatrix} = \begin{bmatrix} -4 \\ 7 \end{bmatrix}.$$

The weights are 2 and -3 ; a negative weight means “go backward along that vector.”

Worked example: combination pills. A pharmacy stocks two combination pills. Pill A contains 5 mg of drug X and 10 mg of drug Y; pill B contains 10 mg of drug X and no drug Y. As vectors of (mg of X, mg of Y),

$$\bar{a} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}, \quad \bar{b} = \begin{bmatrix} 10 \\ 0 \end{bmatrix}.$$

A patient who takes 2 of pill A and 1 of pill B gets $2\bar{a} + 1\bar{b} = \begin{bmatrix} 10 \\ 20 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \end{bmatrix} = \begin{bmatrix} 20 \\ 20 \end{bmatrix}$: 20 mg of each drug. The weights are the numbers of pills, and the linear combination is the total dose. Section 5.3 runs this backward: given a prescribed total, how many of each pill?

Worked example: a day of meals. Suppose one serving of oatmeal has 5, 27, and 3 grams of (protein, carbohydrate, fat) and one serving of yogurt has 10, 7, and 0. Then 2 servings of oatmeal and 1.5 of yogurt give

$$2 \begin{bmatrix} 5 \\ 27 \\ 3 \end{bmatrix} + 1.5 \begin{bmatrix} 10 \\ 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 10 \\ 54 \\ 6 \end{bmatrix} + \begin{bmatrix} 15 \\ 10.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 64.5 \\ 6 \end{bmatrix}.$$

Every meal plan is a linear combination of foods, with servings as the weights.

Watch out: the mathematics allows any weights

In a linear combination the weights can be any real numbers: negative, fractional, huge. The mathematics does not know that you cannot take -0.5 of a pill or eat a negative serving. When a problem has real-world limits like these, check the weights you find against them; a recipe can be mathematically correct and still impossible to carry out (Exercise 5.3 is an example).

5.2 Span: Everything You Can Reach

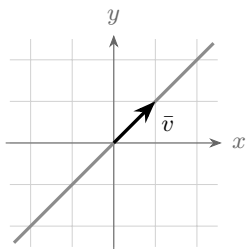
Definition: Span

The **span** of a set of vectors is the set of **all** their linear combinations.

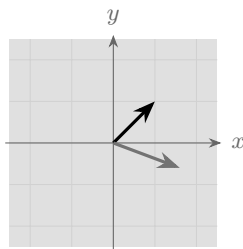
The span answers the first question: it is everything you can reach. In the plane there are three possibilities.

- **One nonzero vector:** its span is a **line** through the origin, since its multiples all lie on one line (Section 3.3).
- **Two, in different directions:** the **whole plane**. Go any amount along the first, then any amount along the second, and you can reach any point.
- **Two on the same line:** only **that line**. The second vector is a multiple of the first and adds no new direction.

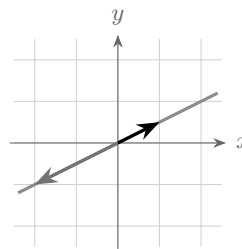
Every span contains $\bar{0}$: set all the weights to zero.



One vector: a line.



Two directions: the plane.



Two on one line: still a line.

In the running examples:

- **Velocity:** taxiing, a plane can only roll forward or back along the runway, so its velocities fill **one line**; flying, it can head any way it likes, so its velocities fill the **whole plane**. (Once it can also climb and descend, they fill all of three-dimensional space.)
- **Data:** the span of your features is every pattern they can express. If every column of a dataset lies on one line, the dataset can only express one pattern, however many columns it has.

Worked example: testing whether a point is on a line. Is $\begin{bmatrix} 6 \\ 4 \end{bmatrix}$ in the span of $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$? The span of one vector is its multiples, so we need one number c with $c \cdot 3 = 6$ and $c \cdot 2 = 4$. Both give $c = 2$: **yes**, $\begin{bmatrix} 6 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Is $\begin{bmatrix} 6 \\ 5 \end{bmatrix}$? The first row still needs $c = 2$, but the second needs $c = 2.5$: no single c works, so **no**. Geometrically, the span of $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$ is the line $y = \frac{2}{3}x$, and $(6, 5)$ is not on it.

Worked example: a span inside $\mathbb{R}^3$. The span of two one-step vectors in $\mathbb{R}^3$ is every vector of the form

$$a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} :$$

all the vectors whose third component is 0. It is the flat “floor” of three-dimensional space, a plane through the origin. So the vector with components $2, -3, 0$ is in the span (take $a = 2, b = -3$), but the vector with components $1, 1, 1$ is not: no choice of a and b can make the third component 1. Two vectors in $\mathbb{R}^3$ can never span all of it; that takes three.

Watch out: every span goes through the origin

Because all-zero weights always give $\bar{0}$, every span contains the origin. A line such as $y = x + 1$, which misses the origin, is *never* a span, however it is drawn. If you sketch a span and it does not pass through $(0, 0)$, something has gone wrong.

Notebook U2-2_Vectors-4_LinearCombinationsAndSpan, §2, the span filled in point by point

5.3 Finding the Weights Is Solving a System

Now the second question: which weights reach a given point? Take

$$\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Is $\bar{u}$ a combination of $\bar{v}$ and $\bar{w}$? Write out what that would mean, with unknown weights a and b :

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a + b = 3 \\ b = 2 \end{cases}$$

Two vectors are equal exactly when their components match, so the vector equation on the left is the same as one ordinary equation per **row**: the top components must agree ($a \cdot 1 + b \cdot 1 = 3$) and so must the bottom ones ($a \cdot 0 + b \cdot 1 = 2$). That is a system of equations, exactly the kind solved in Unit 2-1.

Solve it one row at a time:

- Row 2 gives $b = 2$.
- Row 1 then gives $a = 3 - 2 = 1$.
- So $\bar{u} = 1\bar{v} + 2\bar{w}$.

Check:

$$1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+2 \\ 0+2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Unit 2-1 showed that a system has one solution, no solution, or infinitely many. Each of the three outcomes now has a geometric meaning:

- **One solution:** exactly one recipe reaches $\bar{u}$.
- **No solution:** $\bar{u}$ lies **outside** the span, and no recipe reaches it.
- **Infinitely many:** the vectors are redundant, so many recipes reach the same point.

The worked examples below show all three.

Procedure: finding the weights

1. Write the target as an unknown combination: $a\bar{v} + b\bar{w} = \bar{u}$.
2. Read it one row (component) at a time to get a system of equations.
3. Solve the system by substitution or elimination, as in Unit 2-1.
4. Check by rebuilding the target from the weights you found.

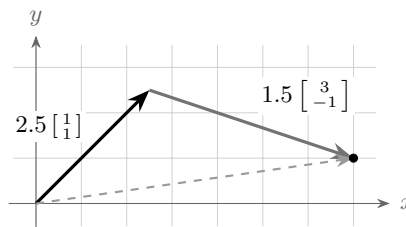
Worked example: one solution, by elimination. Write $\begin{bmatrix} 7 \\ 1 \end{bmatrix}$ as a combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$. In Unit 2-1's bookkeeping the system is the augmented matrix on the right, whose columns are the two vectors and the target:

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a + 3b = 7 \\ a - b = 1 \end{cases} \quad \longleftrightarrow \quad \left[\begin{array}{cc|c} 1 & 3 & 7 \\ 1 & -1 & 1 \end{array} \right]$$

Subtract the second equation from the first: $4b = 6$, so $b = 1.5$. Then the second equation gives $a = 1 + b = 2.5$. Check:

$$2.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1.5 \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2.5 + 4.5 \\ 2.5 - 1.5 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

The picture shows the recipe: two and a half steps along $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then one and a half along $\begin{bmatrix} 3 \\ -1 \end{bmatrix}$.



The recipe, head to tail, reaches $(7, 1)$.

Worked example: the pills, run backward. A patient needs 25 mg of drug X and 30 mg of drug Y. How many of pill A, $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$, and pill B, $\begin{bmatrix} 10 \\ 0 \end{bmatrix}$?

$$a \begin{bmatrix} 5 \\ 10 \end{bmatrix} + b \begin{bmatrix} 10 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 30 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} 5a + 10b = 25 \\ 10a = 30 \end{cases}$$

Row 2 gives $a = 3$; row 1 then gives $15 + 10b = 25$, so $b = 1$. Three of pill A and one of pill B: $3 \begin{bmatrix} 5 \\ 10 \end{bmatrix} + \begin{bmatrix} 10 \\ 0 \end{bmatrix} = \begin{bmatrix} 25 \\ 30 \end{bmatrix}$. Exactly one recipe, because the two pills have different mixes of X and Y (their vectors point in different directions).

Worked example: no solution. Can $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$ be built from $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$?

$$a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a + 2b = 3 \\ 2a + 4b = 5 \end{cases}$$

Doubling the first equation gives $2a + 4b = 6$, but the second says $2a + 4b = 5$. Both cannot hold: **no solution**. The reason is geometric. $\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, so the two vectors lie on one line, the line $y = 2x$, and their span is only that line. The point $(3, 5)$ is not on it ($2 \times 3 = 6 \neq 5$), so it lies outside the span.

Worked example: infinitely many solutions. Build $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ from the three vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a + c = 2 \\ b + c = 3 \end{cases}$$

Two equations, three unknowns. As in Unit 2-1, let the extra unknown be free: choose any value $c = t$, and then $a = 2 - t$ and $b = 3 - t$. Every choice of t is a valid recipe:

t	a	b	c	Check
0	2	3	0	$2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
1	1	2	1	$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
2	0	1	2	$\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

The third vector is redundant: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is already $\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, so any amount of it can be swapped for the same amount of the other two.

5.4 Independence: Is Anything Redundant?

That last outcome, redundancy, has a name.

Definition: Linearly dependent

A set of vectors is **linearly dependent** if one vector is a linear combination of the others. There are two equivalent ways to write this, and **either works**:

$$\bar{u} = a\bar{v} + b\bar{w} \quad \text{or} \quad a\bar{u} + b\bar{v} + c\bar{w} = \bar{0} \quad \text{with } a, b, c \text{ not all } 0.$$

The two forms have different strengths:

- The first names the **redundant** vector, $\bar{u}$.
- The second treats all three vectors alike. To get it from the first, move everything to one side: $1\bar{u} - a\bar{v} - b\bar{w} = \bar{0}$, a combination with weights $1, -a, -b$. The weight on $\bar{u}$ is 1, so the weights are certainly not all zero.
- “Not all 0” matters: all-zero weights give $\bar{0}$ for **any** vectors whatsoever, so finding them proves nothing.

Definition: Linearly independent

A set is **linearly independent** if it is not dependent: **no** vector is a combination of the others. Equivalently, $a\bar{u} + b\bar{v} + c\bar{w} = \bar{0}$ **only** when $a = b = c = 0$.

Independent vectors each contribute a genuinely new direction. There is also a limit on how many a space can hold:

Worth memorizing

In $\mathbb{R}^n$, any **more than** n vectors are dependent.

For example, any three vectors in the plane $\mathbb{R}^2$ are dependent: at least one of them can be built from the others. Section 6 explains why, in terms of dimension.

Worked example: examples and non-examples.

- $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 3 \\ 6 \end{bmatrix}$ are **dependent**: $\begin{bmatrix} 3 \\ 6 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, or in the second form $3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ with weights 3 and -1 , not both zero.
- $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ are **dependent**: $2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \bar{0}$. (Three vectors in $\mathbb{R}^2$ had to be.)
- $\hat{i}$ and $\hat{j}$ are **independent**: $a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$, and that is $\bar{0}$ only if $a = 0$ and $b = 0$.
- Any set containing $\bar{0}$ is **dependent**: $1 \cdot \bar{0} = \bar{0}$ uses a nonzero weight. The zero vector adds no direction at all.

For just **two** vectors the test is quick: they are dependent exactly when one is a multiple of the other, that is, when they lie on one line through the origin.

Watch out: dependent does not mean “each is built from the others”

Dependence needs only *one* vector that is a combination of the others. In the set $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the second is twice the first, so the set is dependent, even though $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ cannot be built from the other two (they both lie on the x -axis). One redundancy is enough to make the whole set dependent.

5.5 Example: Testing for Dependence

Is $\bar{u}$ a combination of $\bar{v}$ and $\bar{w}$? Set up the vector equation and read the components one row at a time:

$$a \underbrace{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}}_{\bar{v}} + b \underbrace{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}_{\bar{w}} = \underbrace{\begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}}_{\bar{u}} \quad \longleftrightarrow \quad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = -1 \end{cases}$$

Solve it:

- Row 1: $a = 2$.
- Row 2: $b = 3 - 2a = 3 - 4 = -1$.
- Row 3: $b = -1$. It **agrees**.

So $\bar{u} = 2\bar{v} - \bar{w}$, and the set is **dependent**: $\bar{u}$ adds nothing new. Check:

$$2 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 - 0 \\ 4 - 1 \\ 0 - 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}.$$

Notice what happened. There were three equations but only two unknowns, so two of the equations were enough to pin down a and b , and the third was a test. The test happened to pass: the equations were **consistent**. That is what “dependent” means here.

5.6 Example: Change One Number and It Flips

Now replace $\bar{u}$ by

$$\bar{u}' = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix},$$

changing only its last component, and ask the same question:

$$a \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a = 2 \\ 2a + b = 3 \\ b = 0 \end{cases}$$

Solve it:

- Rows 1 and 2 are unchanged, so they still force $a = 2$ and $b = -1$.
- Row 3 demands $b = 0$: a **contradiction**.
- So no weights exist, and $\bar{u}'$ is not a combination of $\bar{v}$ and $\bar{w}$. Since $\bar{v}$ and $\bar{w}$ are not multiples of each other either, no vector in the set is built from the others, and the set is **independent**. (Could $\bar{v}$ be built from $\bar{u}'$ and $\bar{w}$ instead? Any such recipe would need some $\bar{u}'$, since $\bar{v}$ is not a multiple of $\bar{w}$; rearranging it would then build $\bar{u}'$ from $\bar{v}$ and $\bar{w}$, which we just saw is impossible. The same argument covers $\bar{w}$.)

The takeaway:

- Testing for dependence **is** solving a system.
- A solution is the redundancy; **no solution** means independent.
- Geometrically, $\bar{v}$ and $\bar{w}$ span a plane inside $\mathbb{R}^3$. The vector $\bar{u}$ lay in that plane; $\bar{u}'$ pokes out of it.

Worked example: testing three vectors at once. The second form of the definition tests a whole set in one go, with no need to guess which vector might be redundant. Are

$$\bar{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \bar{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \bar{v}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

independent? Ask which weights give $\bar{0}$:

$$a\bar{v}_1 + b\bar{v}_2 + c\bar{v}_3 = \bar{0} \quad \longleftrightarrow \quad \begin{cases} a + c = 0 \\ a + b = 0 \\ b + c = 0 \end{cases}$$

Row 1 gives $a = -c$. Row 2 then gives $b = -a = c$. Row 3 becomes $c + c = 0$, so $c = 0$, and then $a = 0$ and $b = 0$. The *only* weights that give $\bar{0}$ are all zero, so the set is **independent**.

Now change $\bar{v}_3$ to $\bar{v}'_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. The system becomes $a + c = 0$, $a + b + 2c = 0$, $b + c = 0$. Rows 1 and 3 give $a = -c$ and $b = -c$; row 2 becomes $-c - c + 2c = 0$, which is true for *every* c . So c is free and there are infinitely many solutions; $c = -1$ gives $a = 1$, $b = 1$:

$$1\bar{v}_1 + 1\bar{v}_2 - 1\bar{v}'_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \bar{0}.$$

The set is **dependent**, and the solution names the redundancy: $\bar{v}'_3 = \bar{v}_1 + \bar{v}_2$. In this form, “only the zero solution” means independent, and “a nonzero solution” means dependent.

5.7 Why Dependence Is a Problem

A spare vector sounds harmless. It is not.

- **No unique answer.** If $\bar{u} = 2\bar{v} - \bar{w}$, then every target reachable from $\{\bar{u}, \bar{v}, \bar{w}\}$ has **infinitely many** recipes, because any amount of $\bar{u}$ in a recipe can be traded for the same amount of $2\bar{v} - \bar{w}$.
- **Singular matrices.** Dependent columns make a matrix (next unit) with no inverse, and `np.linalg.solve` refuses to solve with it.
- **Nearly dependent is almost as bad.** Tiny changes in the data swing the weights wildly, and in floating point (Unit 1-5) you get huge coefficients, **inf**, or **NaN**.

Worked example: two recipes for one target. Use the vectors of the testing example, where $\bar{u} = 2\bar{v} - \bar{w}$, and aim for the target with components 3, 5, -1. Here are two different recipes:

$$1\bar{u} + 1\bar{v} + 0\bar{w} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix},$$

$$0\bar{u} + 3\bar{v} - 1\bar{w} = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}.$$

The second came from the first by trading the $\bar{u}$ for $2\bar{v} - \bar{w}$. Asked “how much does each vector contribute?”, the honest answer is “it depends which recipe you pick,” and that is exactly the trouble a regression gets into with redundant columns.

Worked example: a small change, a big swing. The vectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1.001 \end{bmatrix}$ are not multiples of each other, but they very nearly are. Find weights for two targets that differ by only 0.001:

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1.001 \end{bmatrix} = \begin{bmatrix} 2 \\ 2.001 \end{bmatrix}: \text{ subtracting row 1 from row 2 gives } 0.001b = 0.001, \text{ so } b = 1, a = 1;$$

$$a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 1.001 \end{bmatrix} = \begin{bmatrix} 2 \\ 2.002 \end{bmatrix}: \text{ subtracting gives } 0.001b = 0.002, \text{ so } b = 2, a = 0.$$

A change of one part in two thousand in the target moved the weights from (1, 1) to (0, 2). With real, slightly noisy measurements, weights like these are close to meaningless.

Running example: collinear patient features.

- Suppose you record each patient’s height in inches **and** in centimetres. Then one column of the data table is 2.54 times the other. For three patients of 60, 65, and 70 inches,

$$\text{cm column} = 2.54 \begin{bmatrix} 60 \\ 65 \\ 70 \end{bmatrix} = \begin{bmatrix} 152.4 \\ 165.1 \\ 177.8 \end{bmatrix}.$$

The two columns are dependent.

- A regression (a model that predicts an outcome from the features) cannot decide which of the two deserves the credit, so its coefficients go unstable.
- The fix is to drop or combine the redundant column; PCA automates that.

Worked example: a dependence hiding in a lipid panel. Redundancy is not always as obvious as inches and centimetres. Many labs do not measure LDL cholesterol directly; they *calculate* it from the other three numbers of a lipid panel with the Friedewald formula, $\text{LDL} = \text{TC} - \text{HDL} - \frac{1}{5} \text{TG}$ (total cholesterol, HDL, and triglycerides, all in mg/dL). For three patients:

$$\underbrace{\begin{bmatrix} 200 \\ 180 \\ 240 \end{bmatrix}}_{\text{TC}} - \underbrace{\begin{bmatrix} 50 \\ 60 \\ 40 \end{bmatrix}}_{\text{HDL}} - \frac{1}{5} \underbrace{\begin{bmatrix} 150 \\ 100 \\ 200 \end{bmatrix}}_{\text{TG}} = \begin{bmatrix} 200 - 50 - 30 \\ 180 - 60 - 20 \\ 240 - 40 - 40 \end{bmatrix} = \underbrace{\begin{bmatrix} 120 \\ 100 \\ 160 \end{bmatrix}}_{\text{LDL}}.$$

So the LDL column is exactly the linear combination $1 \text{ TC} - 1 \text{ HDL} - 0.2 \text{ TG}$ of the other columns, for every patient: a dataset holding all four columns is dependent. A model given all four will struggle, and no plot of any two columns would reveal why.

5.8 More Practice: Combinations, Span, and Independence

Exercise 5.1

Is the first vector below a linear combination of the other two? If so, find the weights.

$$\begin{bmatrix} 4 \\ -1 \\ 7 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

Solution to Exercise 5.1

Write the unknown combination and read it row by row:

$$a \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + b \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 7 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} a + 2b = 4 \\ -b = -1 \\ 2a + 3b = 7 \end{cases}$$

Row 2 gives $b = 1$; row 1 gives $a = 4 - 2 = 2$. Row 3 is the test: $2(2) + 3(1) = 7$. It passes, so **yes**, with weights 2 and 1:

$$2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \\ 7 \end{bmatrix}.$$

The target lies in the plane spanned by the two vectors.

Exercise 5.2

Are these three vectors independent? If not, give weights, not all zero, that combine them to $\bar{0}$.

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Solution to Exercise 5.2

Dependent. The second vector is twice the first, so

$$2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - 1 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \bar{0}, \quad \text{with weights } 2, -1, 0.$$

The zero weight on the third vector is fine: “not all zero” only needs one nonzero weight, and here there are two. (Compare the Watch out box in Section 5.4: the third vector is not built from the others, but the set is dependent anyway.)

Exercise 5.3

With pill A = $\begin{bmatrix} 5 \\ 10 \end{bmatrix}$ and pill B = $\begin{bmatrix} 10 \\ 0 \end{bmatrix}$ (mg of drugs X and Y), can a patient get exactly 15 mg of X and 40 mg of Y? Is the target in the span? Can it actually be taken?

Solution to Exercise 5.3

Solve $a \begin{bmatrix} 5 \\ 10 \end{bmatrix} + b \begin{bmatrix} 10 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 \\ 40 \end{bmatrix}$: row 2 says $10a = 40$, so $a = 4$; row 1 says $20 + 10b = 15$, so $b = -0.5$. The target **is** in the span (two pills with different mixes span the whole plane, so every target is), and the recipe is 4 of pill A and -0.5 of pill B. But a patient cannot take -0.5 of a pill, so the target **cannot** be taken with these pills. Four of pill A already deliver 20 mg of X, more than the 15 mg wanted, and there is no pill that removes X. The mathematics answers “is it reachable with any weights?”; whether the weights are allowed is a separate, real-world question.

6 Basis and Change of Basis

6.1 Basis and Dimension

Put span and independence together and you get the right set of building blocks for a space: enough vectors to reach everything, and not one more.

Definition: Basis and dimension

A **basis** is a set of vectors that is **independent** (nothing redundant) and **spanning** (reaches everything). The **dimension** of the space is the number of vectors in a basis.

What to know:

- A basis is **minimal and complete**: drop one vector and you miss something; add one and it is redundant. This is the idea behind the rule of Section 5.4 that more than n vectors in $\mathbb{R}^n$ must be dependent: n vectors are already enough to reach everything, so there is no room for another genuinely new direction.
- A basis is **not unique**. $\hat{i}, \hat{j}$ is the **standard** basis of the plane, but $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ works too: neither is a multiple of the other, so they point in different directions and span the whole plane.
- An **orthonormal** basis is the nicest kind: its vectors are like the standard axes, rotated.

In the running examples:

- **Velocity**: the basis is east and north. Dimension 2 (3 once the plane climbs, adding “up”).
- **Patient**: one axis per measurement. Dimension 3.
- **Email**: one axis per vocabulary word. The vocabulary size **is** the dimension.

Why insist on both conditions? **Spanning** guarantees that every vector has at least one recipe in terms of the basis. **Independence** guarantees that it has only one: if there were two different recipes for the same vector, subtracting one from the other would give a combination equal to $\vec{0}$ with weights not all zero, which independence forbids. So in a basis, every vector has exactly one recipe, and those unique weights are its **coordinates**. That is what makes a basis usable as a set of axes.

Worked example: basis or not.

- $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ in $\mathbb{R}^2$: **a basis**. Two vectors, not multiples of each other ($3/1 \neq 4/2$), so they are independent and span the plane.
- $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix}$: **not a basis**. The second is twice the first; they span only a line.
- $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ alone: **not a basis** of $\mathbb{R}^2$. Independent, but it spans only the x -axis.
- $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$: **not a basis**. It spans the plane, but three vectors in $\mathbb{R}^2$ are dependent (Section 5.3 found infinitely many recipes with exactly these three).
- $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ in $\mathbb{R}^3$: **not a basis** of $\mathbb{R}^3$. They span only the floor. They *are* a basis of that floor, which is a two-dimensional plane sitting inside three-dimensional space.

Watch out: the right number is necessary, not sufficient

Every basis of $\mathbb{R}^n$ has exactly n vectors, so a set of the wrong size can be ruled out at a glance. But n vectors are not automatically a basis: $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ are two vectors in $\mathbb{R}^2$ and still fail, because they are dependent. With the right number of vectors, check independence; if it holds, spanning comes with it.

6.2 Change of Basis: the Same Point, Different Coordinates

The point $(2, 4)$ means $2\hat{i} + 4\hat{j}$. Coordinates are always **amounts of the basis vectors**, so if we change the basis, the same point gets new coordinates.

Take new axes x' and y' , rotated from the standard ones but still orthonormal:

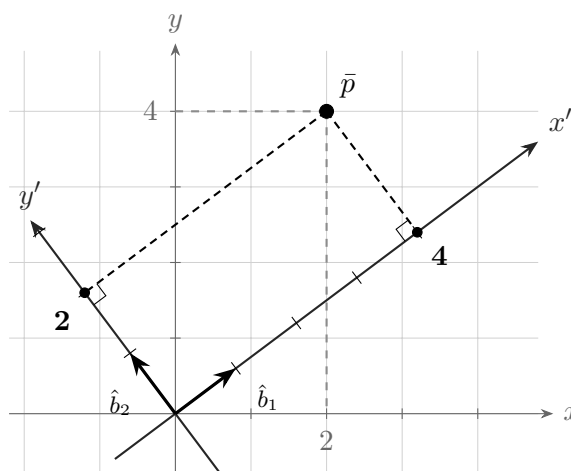
$$\hat{b}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, \quad \hat{b}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$$

Each has length $\sqrt{0.64 + 0.36} = 1$, and $\hat{b}_1 \cdot \hat{b}_2 = -0.48 + 0.48 = 0$.

Each new coordinate is the **projection** of $\bar{p} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$ onto one of the new axes. Because the axis vectors have length 1, that projection's length is just a dot product (Section 4.13):

$$c_1 = \bar{p} \cdot \hat{b}_1 = (2)(0.8) + (4)(0.6) = 4$$

$$c_2 = \bar{p} \cdot \hat{b}_2 = (2)(-0.6) + (4)(0.8) = 2$$



Gray dashes read the xy axes: $(2, 4)$.

Black dashes read the $x'y'$ axes: $(4, 2)$.

Check by rebuilding the point from its new coordinates:

$$4\hat{b}_1 + 2\hat{b}_2 = \begin{bmatrix} 3.2 \\ 2.4 \end{bmatrix} + \begin{bmatrix} -1.2 \\ 1.6 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Same point, new address. In the old basis it is $(2, 4)$; in the new basis it is $(4, 2)$.

The dot-product shortcut depends on the new axes being orthonormal. If the axes are not orthogonal, the projections no longer add up to the point, and you must instead solve $c_1\bar{b}_1 + c_2\bar{b}_2 = \bar{p}$ for the amounts, a system of equations as in Section 5.3.

Notebook U2-2_Vectors-5_IndependenceAndBasis, §4, the same point read off two grids

Worked example: a second point on the same rotated axes. Find the coordinates of $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$ in the basis $\hat{b}_1, \hat{b}_2$:

$$c_1 = (5)(0.8) + (0)(0.6) = 4, \quad c_2 = (5)(-0.6) + (0)(0.8) = -3.$$

Check: $4\begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} - 3\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix} = \begin{bmatrix} 3.2+1.8 \\ 2.4-2.4 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$. The negative coordinate says the point lies on the negative side of the y' axis, which makes sense: $(5, 0)$ sits below the tilted x' axis.

Worked example: the pilot's point of view. The plane's own natural axes are “straight ahead” and “to the left,” not east and north. Take the heading $\hat{h}_1 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$ and the perpendicular direction $\hat{h}_2 = \begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix}$ (swap and change a sign; this is the plane's left). They are orthonormal, so every coordinate is a dot product.

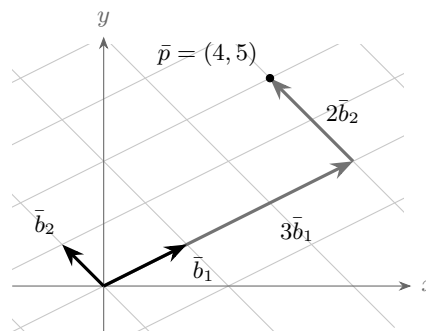
- The plane's own velocity $\begin{bmatrix} 150 \\ 200 \end{bmatrix}$: $c_1 = 90 + 160 = 250$ and $c_2 = -120 + 120 = 0$. In the pilot's axes the velocity is $(250, 0)$: all straight ahead, nothing sideways, as it should be.
- The wind $\bar{w}_1 = \begin{bmatrix} 30 \\ -40 \end{bmatrix}$: $c_1 = 18 - 32 = -14$ and $c_2 = -24 - 24 = -48$. In the pilot's axes the wind is $(-14, -48)$: a 14 m/s headwind and a 48 m/s push to the right (negative “left”).

These are the numbers the projection example of Section 4.13 found. Changing to the pilot's basis *is* projecting onto each of the pilot's axes, and it turns an east-north description into the one that matters in the cockpit.

Worked example: axes that are not orthogonal. The basis $\bar{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ from Section 6 is not orthogonal: $\bar{b}_1 \cdot \bar{b}_2 = -2 + 1 = -1$. Find the coordinates of $\bar{p} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ by solving for the amounts:

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \iff \begin{cases} 2c_1 - c_2 = 4 \\ c_1 + c_2 = 5 \end{cases}$$

Adding the equations eliminates c_2 : $3c_1 = 9$, so $c_1 = 3$ and $c_2 = 5 - 3 = 2$. Check: $3\begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6-2 \\ 3+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$. In this basis the point is $(3, 2)$: three steps along $\bar{b}_1$, then two along $\bar{b}_2$, on the slanted grid in the picture.



The slanted grid of $\bar{b}_1, \bar{b}_2$.

What if we had used the dot-product shortcut anyway? Projecting onto $\bar{b}_1$ gives $\frac{\bar{p} \cdot \bar{b}_1}{\bar{b}_1 \cdot \bar{b}_1} = \frac{8+5}{5} = 2.6$, not 3. The shortcut silently gives the wrong answer, because on slanted axes the perpendicular drops no longer land on the grid lines. Nothing in the arithmetic warns you; you have to know the basis is orthogonal before using it.

Watch out: which method for which basis

- **Orthonormal basis:** each coordinate is a dot product, $c_i = \bar{p} \cdot \hat{b}_i$.
- **Orthogonal, but not unit length:** each coordinate is a projection factor, $c_i = \frac{\bar{p} \cdot \bar{b}_i}{\bar{b}_i \cdot \bar{b}_i}$ (the square-root-free projection of Section 4.13).
- **Not orthogonal:** solve the system $c_1 \bar{b}_1 + c_2 \bar{b}_2 = \bar{p}$. This method always works, for every basis, so it is the safe default.

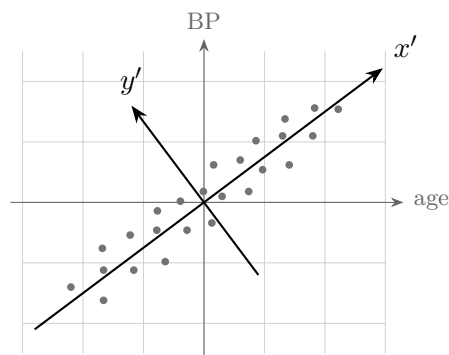
Check the basis first ($\bar{b}_1 \cdot \bar{b}_2 = 0?$), then choose.

6.3 Why Anyone Would Want a Different Basis

The standard basis is convenient, so why change it? Because a well-chosen basis can make data simpler.

Running example: patients.

- Age and blood pressure tend to rise **together**, so a scatter of patients lies along a diagonal.
- Line the axes up with the cloud: the new axis x' carries nearly all the variation, and y' very little.
- **Drop y'** with little loss: two columns become one.



Standardized patients; rotated axes along the cloud.

Worked example: four patients in the new axes. The axes in the picture are the rotated basis of Section 6.2: $\hat{b}_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$ along the cloud (x') and $\hat{b}_2 = \begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}$ across it (y'). Take four standardized patients (age and blood pressure each measured in “typical variations” from average, as in Section 3.5) and find their new coordinates by dot products:

Patient (age, BP)	$x' = \text{patient} \cdot \hat{b}_1$	$y' = \text{patient} \cdot \hat{b}_2$
(1.0, 0.8)	$0.8 + 0.48 = 1.28$	$-0.6 + 0.64 = 0.04$
(-1.5, -1.0)	$-1.2 - 0.6 = -1.8$	$0.9 - 0.8 = 0.1$
(0.5, 0.5)	$0.4 + 0.3 = 0.7$	$-0.3 + 0.4 = 0.1$
(2.0, 1.4)	$1.6 + 0.84 = 2.44$	$-1.2 + 1.12 = -0.08$

The x' coordinates spread from -1.8 to 2.44 ; the y' coordinates never stray more than 0.1 from zero. Almost all of the information is in x' . Now drop y' and rebuild the first patient from x' alone:

$$1.28 \hat{b}_1 = 1.28 \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 1.024 \\ 0.768 \end{bmatrix} \quad \text{versus the true} \quad \begin{bmatrix} 1.0 \\ 0.8 \end{bmatrix}.$$

The error is $\begin{bmatrix} -0.024 \\ 0.032 \end{bmatrix}$, of length 0.04: exactly the dropped y' coordinate, and small. One number per patient now does the work of two, at a cost we can measure.

That is PCA.

- **Principal component analysis**, this unit's capstone, is a change of basis **chosen well**.
- Which basis? **Eigenvectors** answer that, in Unit 2-6.
- It is the same idea behind compressing images and audio: describe the data in axes where most of the numbers are small enough to drop.

6.4 Exercise: Span, Independence, and Basis

Exercise 6.1

1. Is $\begin{bmatrix} 5 \\ 5 \end{bmatrix}$ in the span of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$? Is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$?
2. Are $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$ independent? What do they span?
3. Are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ independent? Answer before computing.
4. Write $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$ in the basis $\bar{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

Solution to Exercise 6.1

1. **Yes** for the first: $\begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. **No** for the second. The span of one vector is the set of its multiples, here the line $y = x$, and the point $(1, 2)$ is not on it. In equations, $c \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ would need $c = 1$ from the top row and $c = 2$ from the bottom row at once, which is impossible.
2. **Dependent**, since $\begin{bmatrix} 2 \\ 6 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix}$. The second vector adds no new direction, so together they span only the line through the origin in the direction $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ (the line $y = 3x$), not the plane.
3. **Dependent**, and no computation is needed: three vectors in $\mathbb{R}^2$ are always dependent, because a basis for $\mathbb{R}^2$ has only two vectors. Here the first two are not multiples of each other, so they already span the plane, and the third has to be a combination of them. (If you want to confirm it, solving $a + b = 2$ and $a - b = 3$ gives $a = 2.5$ and $b = -0.5$, and indeed $2.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.5 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.)
4. The coordinates are the weights c_1, c_2 that rebuild the vector:

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix} \quad \longleftrightarrow \quad \begin{cases} c_1 + c_2 = 4 \\ c_1 - c_2 = 0 \end{cases}$$

Adding the two equations gives $2c_1 = 4$, so $c_1 = 2$, and then $c_2 = 2$ as well. The coordinates are $(2, 2)$ in the $\bar{b}$ basis. Check: $2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$. Note that $\bar{b}_1 \cdot \bar{b}_2 =$

$1 - 1 = 0$, so this basis is orthogonal: its axes are the standard axes turned through 45° , although each basis vector has length $\sqrt{2}$ rather than 1. Because it is orthogonal, the projection factors give the same answer without solving anything. Writing $\bar{t} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$ for the target,

$$c_1 = \frac{\bar{t} \cdot \bar{b}_1}{\bar{b}_1 \cdot \bar{b}_1} = \frac{4 + 0}{2} = 2, \quad c_2 = \frac{\bar{t} \cdot \bar{b}_2}{\bar{b}_2 \cdot \bar{b}_2} = \frac{4 - 0}{2} = 2.$$

6.5 More Practice: Basis and Coordinates

Exercise 6.2

Is this set a basis of $\mathbb{R}^3$?

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Solution to Exercise 6.2

There are three vectors in $\mathbb{R}^3$, the right number, so it comes down to independence. Section 5 tested exactly this set (in the worked example “testing three vectors at once”) and found that the only weights giving $\bar{0}$ are all zero. So the set is independent, and three independent vectors in $\mathbb{R}^3$ span it: **yes, a basis**. Every vector in $\mathbb{R}^3$ has exactly one recipe in terms of these three.

Exercise 6.3

Find the coordinates of $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ in the pilot’s basis $\hat{h}_1 = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$, $\hat{h}_2 = \begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix}$, and check your answer.

Solution to Exercise 6.3

The basis is orthonormal (each has length 1, and $\hat{h}_1 \cdot \hat{h}_2 = -0.48 + 0.48 = 0$), so each coordinate is a dot product: $c_1 = (3)(0.6) + (1)(0.8) = 2.6$ and $c_2 = (3)(-0.8) + (1)(0.6) = -1.8$. Check: $2.6 \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix} - 1.8 \begin{bmatrix} -0.8 \\ 0.6 \end{bmatrix} = \begin{bmatrix} 1.56+1.44 \\ 2.08-1.08 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$.

Exercise 6.4

Find the coordinates of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ in the basis $\bar{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\bar{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, two ways: by solving a system, and with the projection factors.

Solution to Exercise 6.4

By a system: $c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ gives $c_1 + c_2 = 1$ and $c_1 - c_2 = 4$. Adding: $2c_1 = 5$, so $c_1 = 2.5$; then $c_2 = 1 - 2.5 = -1.5$.

By projection factors (allowed, since $\bar{b}_1 \cdot \bar{b}_2 = 0$): $c_1 = \frac{1+4}{2} = 2.5$ and $c_2 = \frac{1-4}{2} = -1.5$.

Both give $(2.5, -1.5)$. Check: $2.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 1.5 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2.5-1.5 \\ 2.5+1.5 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

7 Looking Ahead: Where Matrices Come From

7.1 One System, Four Ways to Write It

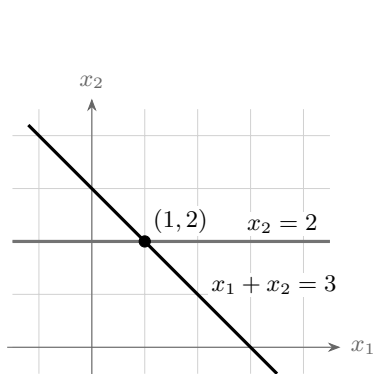
Section 5.3 turned a question about vectors into a system of equations. Run that in reverse and a system of equations becomes a statement about vectors. Here is the same small system in the forms we have met so far, plus the one Unit 2-3 adds. (It is the system from Section 5.3, with the unknowns a and b renamed x_1 and x_2 .)

$$\begin{array}{ccc}
 \begin{array}{c} \text{equations (Unit 2-1)} \\ \left\{ \begin{array}{l} x_1 + x_2 = 3 \\ x_2 = 2 \end{array} \right. \end{array} & \longleftrightarrow & \begin{array}{c} \text{column vectors (this unit)} \\ x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \end{array} \\
 \updownarrow & & \updownarrow \\
 \begin{array}{c} \text{augmented matrix (Unit 2-1)} \\ \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 2 \end{array} \right] \end{array} & \longleftrightarrow & \begin{array}{c} \text{matrix times vector (Unit 2-3)} \\ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad A\bar{x} = \bar{b} \end{array}
 \end{array}$$

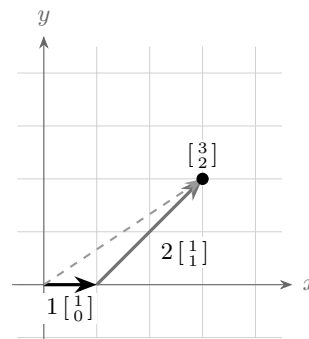
The first three are already familiar. The equations are the system as Unit 2-1 first wrote it; the augmented matrix is the grid of coefficients with the right-hand side attached, which Unit 2-1 used for elimination; and the column-vector form is the linear combination of Section 5. The fourth form is a **preview**: it is new notation, and it is the subject of Unit 2-3. Stand the two columns of the column-vector form side by side and you get a grid of numbers called a **matrix**, A ; collect the unknowns into a vector $\bar{x}$ and the right-hand side into $\bar{b}$, and the whole system fits on one short line, $A\bar{x} = \bar{b}$. For now, read that line as nothing more than shorthand for the column-vector form next to it.

The same system can be read in two directions:

- The **row picture**, read **across**: each equation is a line, and the solution is where the lines cross. Here $x_2 = 2$ is a horizontal line and $x_1 + x_2 = 3$ a slanted one, and they cross at $x_1 = 1$, $x_2 = 2$.
- The **column picture**, read **down**: the unknowns weight the columns, and the solution is the recipe. Here $1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, the same answer found in Section 5.3.



Row picture: two lines cross at the solution.



Column picture: the recipe reaches the right-hand side.

Both pictures find the same two numbers, $x_1 = 1$ and $x_2 = 2$, but they mean different things by them. In the row picture, $(1, 2)$ is a *point* where two lines meet. In the column picture, 1 and 2 are *weights*: how much of each column to take. Linear algebra constantly switches between these two readings, and the matrix notation of Unit 2-3 is built to hold both.

Worked example: the change-of-basis system, three ways. The non-orthogonal basis example of Section 6.2 was also a system in disguise. Finding the coordinates of $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ in the basis $\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ can be written as

$$\begin{cases} 2x_1 - x_2 = 4 \\ x_1 + x_2 = 5 \end{cases} \longleftrightarrow x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \longleftrightarrow \left[\begin{array}{cc|c} 2 & -1 & 4 \\ 1 & 1 & 5 \end{array} \right].$$

The row picture has the lines $2x_1 - x_2 = 4$ and $x_1 + x_2 = 5$ crossing at $(3, 2)$; the column picture takes 3 of the first column and 2 of the second. The coordinates in a new basis are the solution of a system whose columns are the basis vectors.

Where this is going: the columns side by side are a **matrix** A , and Unit 2-3 defines $A\bar{x}$ as those columns weighted by the entries of $\bar{x}$.

Exercise 7.1

Write the system

$$\begin{cases} x_1 + 2x_2 = 5 \\ 3x_1 - x_2 = 1 \end{cases}$$

in column-vector form and as an augmented matrix. Solve it, and check the answer in the column picture.

Solution to Exercise 7.1

Read down the columns: the coefficients of x_1 form the first column, those of x_2 the second.

$$x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}, \quad \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & -1 & 1 \end{array} \right].$$

To solve, the second equation gives $x_2 = 3x_1 - 1$; substituting into the first, $x_1 + 2(3x_1 - 1) = 5$, so $7x_1 = 7$ and $x_1 = 1$, and then $x_2 = 2$. In the column picture, $1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1+4 \\ 3-2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, the right-hand side. The solution is the recipe.

8 Review

What we covered: vectors, operations, angles, and similarity

1. A **vector** has magnitude and direction. We draw it as an arrow and write it as a list of numbers, a **column** by default or a **row**; the two shapes differ, and Unit 2-3 says how.
2. A **point** is a location, a vector is a **displacement**; a vector may be slid anywhere without changing.
3. A vector with n components lives in $\mathbb{R}^n$, and **one row of a data table is one such vector**.

4. **Magnitude** $\|\vec{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}$ is the Pythagorean hypotenuse.
5. **Addition** is head to tail, **scaling** stays on the vector's line, **subtraction** adds the reverse, and $\hat{v} = \vec{v}/\|\vec{v}\|$ is the **unit vector**.
6. **Distance** is $\|\vec{v} - \vec{w}\|$; it is scale-sensitive, so standardize first.
7. The **dot product** $\vec{v} \cdot \vec{w} = \sum v_i w_i$ multiplies matching components and adds; it returns a **number**, not a vector.
8. $\cos \theta = \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}$ gives the **angle**; **cosine similarity** ignores size.

What we covered: projection, span, basis, and the look ahead

9. A dot product of 0 means **orthogonal**, and **projection** is the shadow of one vector on another.
10. A **linear combination** scales and adds; the **span** is everything reachable that way, and finding the weights is **solving a system** from Unit 2-1.
11. **Dependent** vectors are redundant (this is **collinearity**); independent ones each add a new direction.
12. A **basis** is independent and spanning, and its size is the **dimension**; **change of basis** re-describes the same point in different coordinates, which is the idea PCA is built on.
13. A system is also a **linear combination of column vectors**; collecting those columns into a grid gives a **matrix**, the subject of Unit 2-3.

Notebooks:

- U2-2_Vectors-1_ArrowsAndOps.ipynb
- U2-2_Vectors-2_DotProductAndAngle.ipynb
- U2-2_Vectors-3_ProjectionAndSimilarity.ipynb
- U2-2_Vectors-4_LinearCombinationsAndSpan.ipynb
- U2-2_Vectors-5_IndependenceAndBasis.ipynb

Previous: Unit 2-1, **Systems**.

Next: Unit 2-3, **Matrices**: the grid of columns from Section 7 becomes an object in its own right, and multiplying a matrix by a vector gets its definition.